Learning moisture transport parameters of blended concretes from drying curves: identifiability, generalisation across binders, and a comparison of physics-embedded and physics-informed neural networks

Abderrahmane Sghuri[a,b], Xuande Chen[a,b,*], Xiaodong Wang[a]

[a] *Department of Mathematics, Computer Science and Engineering, Université du Québec à Rimouski (UQAR), Rimouski, QC, Canada*

[b] *Research Center on Concrete Infrastructures (CRIB), Québec, QC, Canada*

* *Corresponding author: [e-mail address]*

Highlights

- Drying curves identify only two moisture transport parameters of concrete per mix.
- A solver-embedded network learns them from mix composition with $R^2 = 0.9931$.
- It matches two-stage inversion; flexible composition models fail on new binders.
- Pooling mixes in one model keeps predictions stable when drying curves are short.
- A PINN fits the same data, but its parameters fail an independent solver check.

Abstract

Gravimetric drying tests give the average saturation of concrete specimens, from which moisture transport parameters are usually identified mix by mix. We examine what can be learned across mixes and which modelling choices make the result trustworthy. A vapour–capillary diffusion model of the actual disc specimens was fitted to 932 drying measurements of 20 plain, binary and ternary concretes from a published database, after correcting a time offset between duplicated digitised series. Profile likelihoods showed that only two parameters per mix are identifiable: the reference diffusivity to within 0.03 decade and the intrinsic permeability to within 0.32 decade (median 95% intervals). A physics-embedded neural network (PENN), which maps composition to these parameters and is trained through tabulated solutions of the transport equation, predicted six withheld mixes with $R^2 = 0.9931$. Classical inversion followed by linear regression was equally accurate, and the two did not differ significantly in leave-one-mix-out or leave-one-binder-family-out validation. With a binder family withheld, flexible composition models failed for several mixes (lowest R^2 0.20), whereas the linear physics-based models kept $R^2 \geq 0.962$. End-to-end training was more robust when the training curves were short. Synthetic tests revealed that weight decay compressed the learned permeabilities; a linear map trained to convergence removed this bias. A physics-informed neural network fitted the curves but returned permeabilities about 1.1 decade too low, a consequence of its soft-constrained loss. Fixed constitutive constants shift the parameters by up to 0.4 decade without changing the ranking of the mixes.

Keywords: Concrete; Moisture transport; Drying; Supplementary cementitious materials; Inverse analysis; Identifiability; Physics-embedded neural network; Physics-informed neural network

1 Introduction

The moisture state of concrete controls most of its deterioration processes. Water carries chlorides to the reinforcement, governs the rate of carbonation, drives freeze–thaw damage and alkali–silica reaction, and its loss causes drying shrinkage [1, 2]. Service-life models therefore need moisture transport properties, and these depend strongly on the water-to-binder ratio (w/b) and on supplementary cementitious materials (SCMs) such as fly ash and slag, which refine the pore structure [3, 4]. Moisture moves partly as liquid under capillary pressure gradients and partly as vapour through the gas-filled pores, and the balance between the two changes with the local degree of saturation [5, 6].

The most common laboratory measurement is gravimetric: a specimen is weighed while it dries or absorbs water, which gives its average degree of saturation $S_{\mathrm{avg}}(t)$ [7, 8]. The test is inexpensive, but the curve does not show the moisture profile inside the specimen or the material functions that produce it. These are obtained by inverse analysis: a transport model [6, 9–11] is run repeatedly and its parameters are adjusted until the simulated mass loss matches the measurement. Two difficulties are well known. The problem is ill-posed, because different parameter sets can reproduce the same curve [12, 13], and the calibration is carried out mix by mix, so a new composition needs a new test and a new inversion. Practical identifiability, i.e. whether the available data constrain each parameter, is rarely examined in this context, although tools such as profile likelihoods are well established in other fields [14].

Machine learning has been used to relate composition to transport indicators across many mixes, most often for chloride diffusion coefficients [15]. These models are purely statistical and do not provide the material functions that a transport simulation needs. Physics-informed neural networks (PINNs) offer a way to combine data and governing equations [16–18]. A PINN represents the unknown field by a neural network and adds the residual of the governing equation, computed by automatic differentiation [19], to the data misfit; unknown coefficients are learned in the same optimisation. PINNs have identified constitutive relations of unsaturated soils [20], chloride diffusion coefficients of concrete and alkali-activated mortars [21, 22] and moisture diffusivities of stone [23], and neural networks have recovered moisture diffusivities of concrete from synthetic profiles [24]. Their training is, however, fragile for stiff transport problems: the optimiser can settle on fields that satisfy the residual almost everywhere while missing the initial condition or the early dynamics [25–28]. In an inverse setting this is easily overlooked, because the network may still fit the measurements closely.

An alternative is to keep a conventional solver and place it inside the learning loop, so that gradients of the data misfit flow through the solver to the quantities being learned [29, 30]. Bandai et al. [31] used this differentiable-solver approach to learn soil water retention and conductivity functions from moisture data. The governing equation then holds to the accuracy of the discretisation. For concrete, where the unknowns are a few parameters per mix, the same idea can be combined with a model of how these parameters depend on composition. Whether such end-to-end learning offers anything over the simpler two-stage route — inverting each mix and regressing the results on composition — has not been tested, and neither has the consistency of PINN-identified moisture parameters with an independent solver.

This paper addresses these questions on a published drying database of plain, binary and ternary concretes [8]. Its contributions are:

1. a documented reconstruction of the database, including statistical evidence that its duplicated series are one curve digitised with a time offset;

2. a practical identifiability analysis of a vapour–capillary diffusion model of the real disc specimens, with profile likelihoods that account for correlated residuals, synthetic recovery tests, and the sensitivity of the parameters to fixed constants and boundary conditions;
3. a physics-embedded neural network (PENN) that learns the composition–parameter relationship through tabulated solutions of the transport equation, compared under identical folds with two-stage inversion (four regression models) and a data-driven network, including binder-family extrapolation, mix-level statistics, robustness tests and data-efficiency studies;
4. a controlled comparison with a PINN, with diagnostic runs that locate the origin of its parameter inconsistency.

Section 2 describes the database, Section 3 the transport model and Section 4 the methods. The results are presented in Section 5, discussed in Section 6 and summarised in Section 7.

2 Experimental database

2.1 Source and test protocol

The data come from the experimental programme of Zhuang et al. [8]. Concrete cylinders 100 mm in diameter and 150 mm high were cured sealed at 20 ± 2 °C and 95 ± 5% relative humidity. At the test age, discs 100 mm in diameter and 50 mm thick were cut from the centre of the cylinders, vacuum-saturated for 24 h and dried in an oven at 50 °C, following the conditioning procedure of ASTM C1585 [32]. The discs were weighed daily until the mass loss over 24 h fell below 0.1%, and the degree of saturation was plotted against drying time. The source reports drying curves at curing ages of 28 and 56 days [8]. The curves used here were digitised from the published figures; the curing age of the digitised curves is not recorded in the data file, and the source does not state whether the lateral surface was sealed during drying.

2.2 Concrete mixes

The programme covers four w/b ratios (0.3–0.6) and seven binder systems: plain Portland cement (C), 20% and 40% fly ash (F20, F40), 20% and 40% slag (S20, S40), and ternary blends with 10% + 10% and 20% + 20% fly ash and slag (F10S10, F20S20). The proportions are given in Table 1.

Table 1. Mix proportions of the concretes (kg/m³), after Zhuang et al. [8]. S20-0.5 has no drying curve in the digitised data.

Mix	w/b	Water	Cement	Fly ash	Slag	Used
C-0.3	0.3	179	596	–	–	training
C-0.4	0.4	204	509	–	–	withheld
C-0.5	0.5	226	451	–	–	withheld
C-0.6	0.6	236	393	–	–	–
F20-0.3	0.3	173	461	115	–	withheld
F20-0.4	0.4	200	399	100	–	training
F20-0.5	0.5	222	355	89	–	training
F20-0.6	0.6	233	311	78	–	–
F40-0.3	0.3	167	334	223	–	training
F40-0.4	0.4	197	296	197	–	training
F40-0.5	0.5	217	260	174	–	training
F40-0.6	0.6	231	231	154	–	–
S20-0.3	0.3	176	470	–	118	training
S20-0.4	0.4	205	409	–	102	training
S20-0.5	0.5	227	363	–	91	–
S20-0.6	0.6	237	316	–	79	–
S40-0.3	0.3	174	348	–	232	training
S40-0.4	0.4	205	307	–	205	withheld
S40-0.5	0.5	223	268	–	178	training
S40-0.6	0.6	237	237	–	158	–
F10S10-0.3	0.3	175	466	58	58	withheld
F10S10-0.4	0.4	202	403	50	50	training
F10S10-0.5	0.5	223	357	45	45	training
F10S10-0.6	0.6	236	315	39	39	–
F20S20-0.3	0.3	170	341	114	114	training
F20S20-0.4	0.4	196	295	98	98	withheld
F20S20-0.5	0.5	224	269	90	90	training
F20S20-0.6	0.6	241	241	80	80	–

2.3 Reconstruction of the drying curves

The digitised file contains 1,562 points for 27 mixes. For most mixes it holds two series: a short series with daily points up to 14 days and a long series extending to 40–100 days. Over their common saturation range, the long series is the short series delayed by a nearly constant time (median 1.05 d, interquartile range 0.97–1.13 d over the 18 mixes in which the two agree). Two curves measured on different specimens or at different curing ages would differ by a change of time scale rather than by a shift, since diffusion-controlled drying scales with t/D. Fitting both relations to every mix with w/b ≤ 0.5 settles the question: a pure shift reproduces the short series to within 0.005 in saturation (median RMS), a pure time scaling only to within 0.026, and a free combination of both returns a scale factor of 0.98 (interquartile range 0.97–1.00), i.e. no scaling. The two series are therefore one curve plotted with two time origins; a one-day offset is consistent

with time counted from the start of the 24 h vacuum saturation. Each curve was rebuilt from the short series up to 14 days and the time-corrected long series beyond (Fig. 1a,b). The offset estimated from the training mixes alone (1.06 d) differs from the global value by less than 0.02 d.

The w/b = 0.6 series could not be repaired. The long series of S20-0.6 and S40-0.6 are identical copies, the short series of F20S20-0.6 mixes two curves, and the offsets between the two series are erratic and of the wrong sign (Fig. 1c). The remaining w/b = 0.6 short series dry more slowly at early ages than the corresponding w/b = 0.5 curves, which is implausible, and a classical fit drives the permeability to the lower bound of its range. These mixes were excluded. The database used below contains 932 points for 20 mixes with w/b between 0.3 and 0.5. All steps are scripted, and the effect of the reconstruction choices on the results is quantified in Section 5.5.

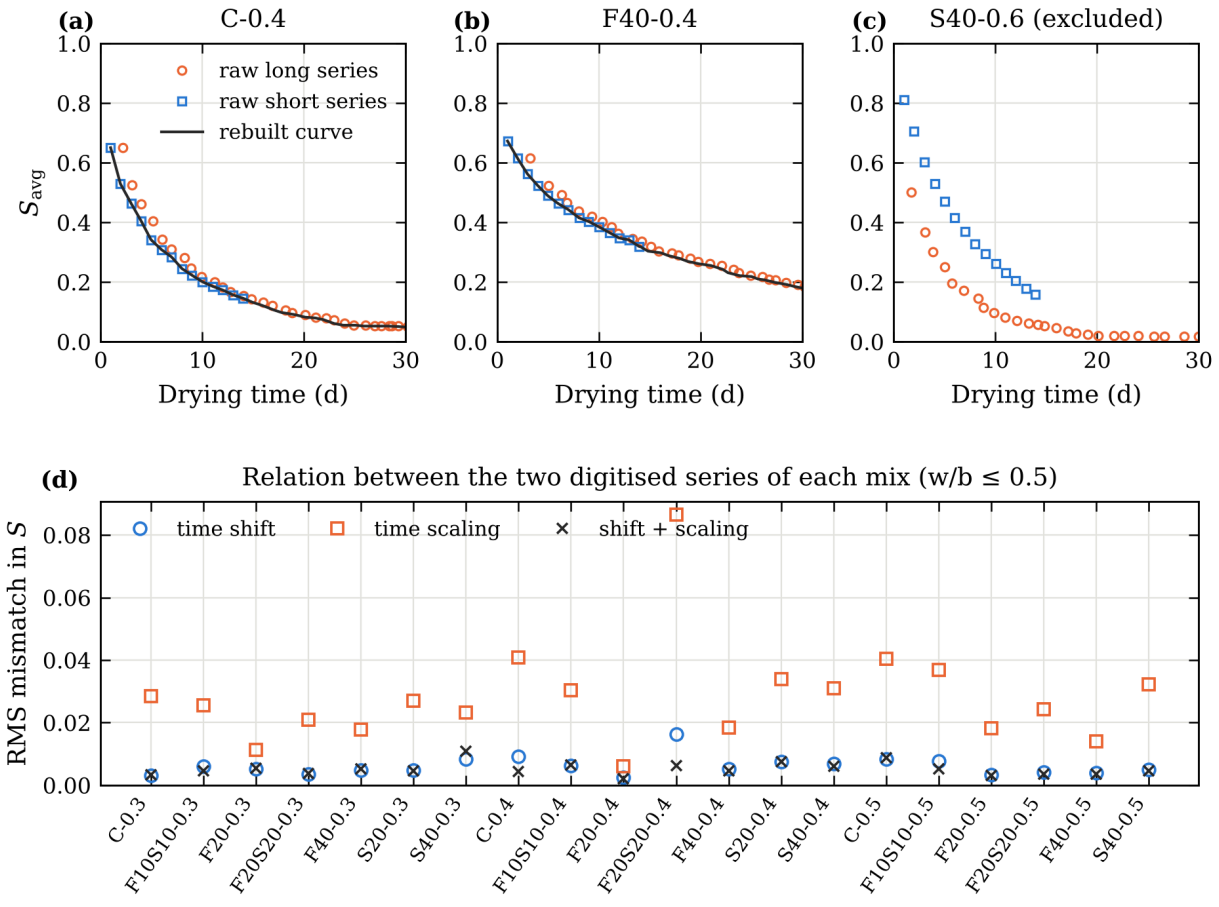


Fig. 1. Reconstruction of the drying curves. (a) C-0.4 and (b) F40-0.4: raw short and long series and the rebuilt curve. (c) S40-0.6: inconsistent series, mix excluded. (d) Median RMS mismatch between the two series of each mix (w/b ≤ 0.5) when they are related by a time shift, a time scaling, or both.

3 Moisture transport model

3.1 Governing equation and diffusivity

Once hydration has largely stopped, sources and sinks of water can be neglected [3], and moisture conservation reads

$$\frac{\partial S}{\partial t} = \nabla \cdot \left[D(S;\boldsymbol{\theta})\, \nabla S \right] \tag{1}$$

where $S \in [0,1]$ is the degree of saturation and D [m²/s] the effective moisture diffusivity. We use the dual-regime diffusivity of Chen et al. [6], which blends a capillary and a vapour contribution:

$$D(S;\boldsymbol{\theta}) = f(S)\, D_{\mathrm{cap}}(S) + \left[1 - f(S)\right] D_{\mathrm{vap}}(S) \tag{2}$$

$$D_{\mathrm{vap}}(S) = D_v \left[\alpha_0 + \frac{1-\alpha_0}{1 + \left(\dfrac{1-S}{1-S_c} \right)^{n}} \right] \tag{3}$$

$$D_{\mathrm{cap}}(S) = \frac{K_{\mathrm{int}}}{\eta_l} \left| \frac{\partial P_c}{\partial S} \right|, \qquad \frac{\partial P_c}{\partial S} = -\beta\, P_{c0}\, S^{-\beta-1} \tag{4}$$

$$f(S) = \frac{1}{1 + \exp\left[-k \left(S - S_{\mathrm{cap}} \right) \right]} \tag{5}$$

Equation (3) has the sigmoidal form of Bažant and Najjar [9], with D_v the reference diffusivity, α_0 the ratio of minimum to maximum vapour diffusivity, n a shape exponent and S_c the transition saturation. In Eq. (4), K_{int} is the intrinsic permeability, η_l the viscosity of water, P_{c0} a reference capillary pressure and β an exponent. The switch (5) makes vapour transport dominant at low saturation and capillary transport dominant at high saturation. The fixed constants are listed in Table 2. Since D_{cap} depends on K_{int} only through $K_{\mathrm{int}} P_{c0} / \eta_l$, only this product is identifiable: a different choice of P_{c0} or η_l shifts $\log_{10} K_{\mathrm{int}}$ by a known constant and leaves every prediction unchanged.

Table 2. Fixed constants of the transport model. Their influence on the identified parameters is quantified in Section 5.9.

Symbol	Meaning	Value	Source
R, H	Disc radius and half-thickness	50 mm, 25 mm	[8]
T	Drying temperature	50 °C	[8]
η_l	Viscosity of water at 50 °C	5.47 × 10⁻⁴ Pa·s	–
P_{c0}	Reference capillary pressure	5 MPa	[6]
β	Capillary pressure exponent	2.5	[6]
S_c	Transition saturation in Eq. (3)	0.80	[6]
S_{cap}, k	Switching saturation and steepness	0.75, 50	[6]
α_0	Minimum-to-maximum diffusivity ratio	0.05	Section 5.2
n	Shape exponent, shared by all mixes	8	Section 5.2

3.2 Specimen geometry and boundary conditions

With the axis and the mid-plane as planes of symmetry, Eq. (1) is solved on $0 \le r \le R$, $0 \le z \le H$ with

$$S(r,z,0) = 1, \qquad S(R,z,t) = S(r,H,t) = 0, \qquad \frac{\partial S}{\partial r}(0,z,t) = \frac{\partial S}{\partial z}(r,0,t) = 0$$

The specimen starts saturated, and the exposed surfaces are assumed to reach the oven-dry state immediately, which defines $S = 0$. The quantity compared with the measurements is the volume average

$$S_{\mathrm{avg}}(t) = \frac{2}{R^2 H} \int_0^H \int_0^R S(r, z, t)\, r \, \mathrm{d}r \, \mathrm{d}z \qquad (7)$$

A one-dimensional slab whose half-thickness equals the volume-to-surface ratio, $L = V/A = 12.5$ mm, is a common simplification. It reproduces the early drying of the disc but not its later stages, because the edges dry from two sides: with the same parameters, the slab curve departs from the disc curve by up to 0.13 in S_{avg}. A slab model fitted to the same data returns parameters biased by −0.33 to −0.25 decade in D_v and up to +0.55 decade in K_{int} (Fig. 2). The disc geometry is therefore used for every result on transport parameters; the slab is kept only for the controlled comparison with the PINN (Section 5.7), where both methods use the same simplified model.

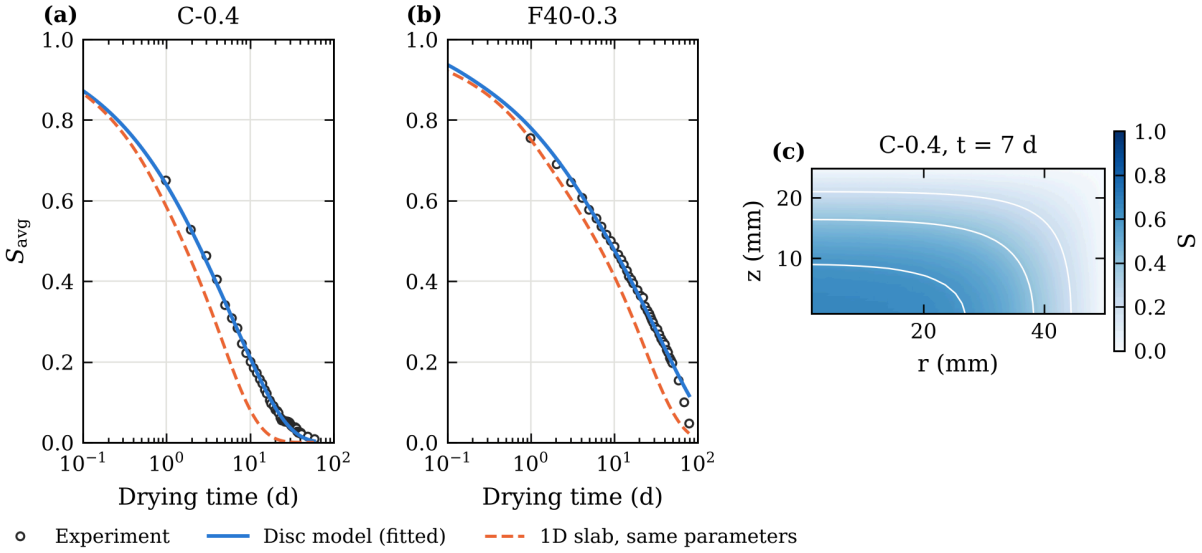


Fig. 2. Disc model and one-dimensional slab. (a, b) Measured average saturation, disc model fitted to the data, and slab model ($L = V/A$) run with the same parameters. (c) Saturation in one quarter of the disc after 7 days for C-0.4; the edge dries from two sides.

4 Methods

4.1 Numerical solution and verification

Equation (1) is discretised with a cell-centred finite-volume scheme on a mesh refined towards the drying faces (24 × 16 cells in (r, z) for the disc, 60–80 cells for the slab) and integrated in time with an implicit BDF scheme [33] at a relative tolerance of 10^{-6}. This direct solver is the reference model of the study. It was verified against the analytical solution for constant diffusivity, by mesh and time-step refinement, by a global water balance, and by checking that a disc of very large radius reproduces the slab solution (Section 5.1, Appendix A).

Learning requires many evaluations of the model and gradients with respect to its parameters. Because only two parameters are identified (Section 5.2), the disc problem was solved once with the direct solver on a 49 × 49 grid of $(\log_{10} D_v, \log_{10} K_{\mathrm{int}})$ covering $[-10, -7.5] \times [-21, -18.5]$, at 120

logarithmically spaced times between 0.05 and 110 days. Between grid points, S_{avg} is interpolated with bicubic convolution [34] in the parameters and linearly in log-time. The interpolant is differentiable and every value it returns is a combination of solutions of Eq. (1). We refer to it as the tabulated solver; it is used for training and, for speed, in the extensive validation experiments, after checking that it reproduces the direct solver to within 9×10^{-5} in S_{avg} and the classical parameter estimates to within 0.002 decade. For the slab, a differentiable version of the finite-volume scheme (linearly implicit Euler, 240 logarithmic time steps) is used instead.

4.2 Classical inversion and identifiability

For each mix, $\boldsymbol{\theta} = [\log_{10} D_v, \log_{10} K_{\mathrm{int}}]$ is estimated by nonlinear least squares (trust-region reflective algorithm, several starting points). Practical identifiability was assessed in three ways. First, the Jacobian at the optimum gives parameter correlations and a condition number, computed for two, three and four free parameters. Second, profile likelihoods [14] were computed for every mix: one parameter is fixed on a grid and the other re-optimised, and the 95% interval is the range where the deviance $N_{\mathrm{eff}} \ln(\mathrm{SSE}/\mathrm{SSE}_{\min})$ stays below 3.84. Digitised curves have strongly autocorrelated residuals, so the number of points N was replaced by $N_{\mathrm{eff}} = N(1-\rho_1)/(1+\rho_1)$, with ρ_1 the lag-1 autocorrelation of the residuals; this widens the intervals. Third, synthetic curves with known parameters, generated on a finer mesh (36 × 24) than the one used for inversion, were inverted with and without perturbations (Section 5.2). The shared exponent n was selected by profiling the total misfit of five training mixes; including or excluding the withheld mix C-0.5 does not change the choice.

4.3 Models linking composition and transport parameters

The composition of a mix is described by $\mathbf{c} = [(\mathrm{w/b} - 0.3)/0.3,\ \mathrm{FA}/0.4,\ \mathrm{SL}/0.4]$, where FA and SL are the fly ash and slag mass fractions of the binder. Two families of models predict $\boldsymbol{\theta}(\mathbf{c})$ for a mix that was not tested.

Two-stage inversion. The classical estimates of the training mixes are regressed on $\mathbf{c}$, and the transport model is run with the regressed parameters. Four regressions were compared: ordinary least squares (OLS, linear), ridge regression with the penalty selected by leave-one-out cross-validation on the training mixes, quadratic features with the same ridge penalty, and Gaussian-process (GP) regression with an anisotropic squared-exponential kernel and a noise term, fitted by maximum marginal likelihood [35].

Physics-embedded neural network. In the PENN (Fig. 3), a small network $g_{\mathbf{w}}$ maps $\mathbf{c}$ to the parameters, mapped smoothly into the admissible range,

$$\boldsymbol{\theta}(\mathbf{c}) = \boldsymbol{\theta}_{\min} + (\boldsymbol{\theta}_{\max} - \boldsymbol{\theta}_{\min}) \odot \sigma\big(g_{\mathbf{w}}(\mathbf{c})\big) \tag{8}$$

with σ the logistic function. The parameters are passed to the tabulated solver and the weights minimise the misfit over all training points,

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{j=1}^{N} \left[S_{\mathrm{avg}}(t_j; \boldsymbol{\theta}(\mathbf{c}_j)) - S_{\mathrm{avg,exp}}(t_j) \right]^2 \tag{9}$$

with gradients obtained by automatic differentiation through the solver [29, 30]. No per-mix inversion is needed: the network is trained directly on the drying curves of all training mixes at once. The reference PENN has two hidden layers of 16 tanh units; a reduced network (one layer of 8 units) and a linear map ($g_{\mathbf{w}}$ affine in $\mathbf{c}$) were also tested. Training uses Adam [36] (learning rate 0.02, decay 0.995 per iteration, weight decay 10^{-4}) for 400 full-batch iterations, after which the training error no longer changes (Fig. D.1a). A revised PENN, introduced in Section 5.6, uses a linear map without weight decay trained to convergence with full-batch L-BFGS.

Data-driven network. A network with four hidden layers of 48 tanh units maps $\left(\sqrt{t/t_{\mathrm{ref}}}, \mathbf{c}\right)$, $t_{\mathrm{ref}} = 100$ d, directly to S_{avg} (Adam, 4000 iterations).

The hyper-parameters of the PENN and of the data-driven network were checked by an inner leave-one-family-out cross-validation restricted to the training mixes (Appendix C), so that no setting was chosen with information from the withheld mixes.

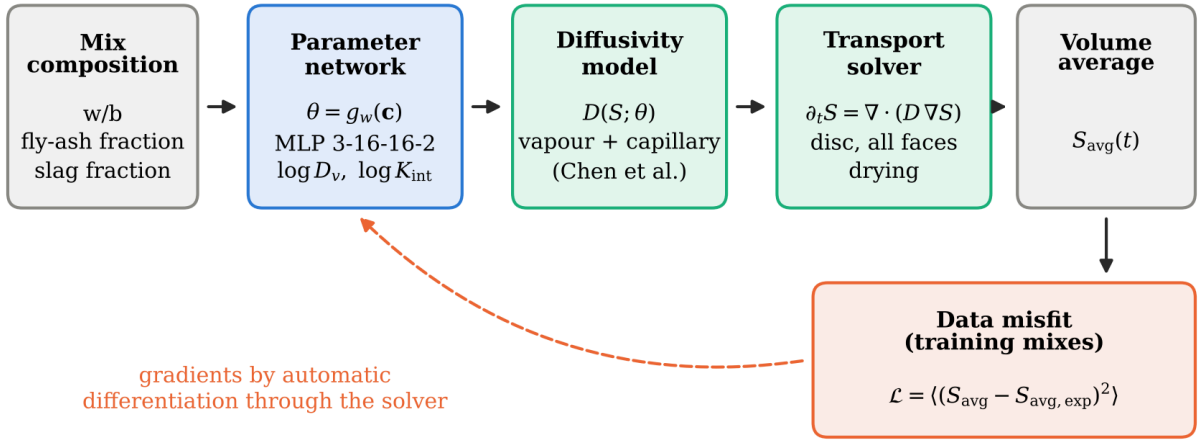


Fig. 3. Physics-embedded neural network. The parameter network predicts the transport parameters from the composition; the tabulated solver returns the average saturation; the data misfit is back-propagated through the solver to the network weights. In the two-stage route, the parameters are instead identified mix by mix and then regressed on composition.

4.4 Physics-informed neural network

The PINN was set up on the slab version of the problem ($L = 12.5$ mm, $n = 2$ from the corresponding profile) with the same parameter network (8). A field network N with four hidden layers of 48 tanh units takes $(\xi, s, \mathbf{c})$ as input, with $\xi = x/L$ and $s = \sqrt{t/t_{\mathrm{ref}}}$. The boundary condition at the drying face and the initial condition are built into the field,

$$S = 1 - \exp\left(-\frac{\xi}{s} A\right), \qquad A = a(s) + \mathrm{softplus}(N) \tag{10}$$

so that $S \to 1$ when $t \to 0$ and $S = 0$ at $\xi = 0$. The loss combines the integral data misfit, the dimensionless PDE residual and the symmetry condition at $\xi = 1$,

$$\mathcal{L} = \sigma_d^{-2}\, \mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{PDE}}\, \mathcal{L}_{\mathrm{PDE}} + \lambda_{\mathrm{sym}}\, \mathcal{L}_{\mathrm{sym}} \tag{11}$$

with $\sigma_d = 0.01$, $\lambda_{\mathrm{PDE}} = 10$ and $\lambda_{\mathrm{sym}} = 1$. The residual is multiplied by t/t_{ref}, which keeps it bounded at early times, and half of the 3000 collocation points per iteration are sampled in the similarity variable ξ/s so that the drying front is resolved at every time. Training uses 3000 Adam iterations followed by 1000 L-BFGS iterations (Fig. D.1b). Without the term $a(s) = 10^{-3} + 0.1\exp[-(s/0.2)^2]$, which bounds the speed of the drying front at early times, the PINN converged in every trial to a trivial field that dries almost instantly, a known failure mode for stiff transport problems [25, 27] that causal weighting of the residual [27] did not remove.

4.5 Validation protocol

1. **Hold-out test.** Six mixes covering four binder families and three w/b ratios (C-0.4, C-0.5, F10S10-0.3, F20-0.3, F20S20-0.4 and S40-0.4; 278 points) were withheld; the other 14 mixes (654 points) were used for training. Networks were trained with five random seeds (three for the PINN).
2. **Cross-validation.** Leave-one-mix-out (LOMO, 20 folds) and leave-one-binder-family-out (LOFO, 7 folds), one seed per fold. Because the points of one curve are not independent, the mix is the statistical unit: methods are compared through per-mix RMSE differences, with 95% bootstrap intervals (10 000 resamples of the mixes [37]) and Wilcoxon signed-rank tests.
3. **Robustness.** The training data were perturbed by independent noise (standard deviation 0.01 and 0.02), by correlated digitisation-like errors (per-curve offset plus AR(1) noise), by random time-origin errors (0.25 d), and by alternative reconstructions of the curves (offset at the bounds of its interquartile range, short or long series only). Predictions were always scored against the unperturbed withheld curves.
4. **Data efficiency.** Models were trained on 5, 8, 11 or 14 randomly drawn mixes (10 draws each, identical for all methods) and tested on the remaining ones, and on training curves truncated after 3, 4, 7 or 14 days.
5. **Solver check.** The parameters returned by each model were inserted, without fitting, into the direct solver, and the resulting curves were compared with the model output.

Accuracy is reported as the coefficient of determination and the root-mean-square error,

$$R^2 = 1 - \frac{\sum_j (y_j - \hat{y}_j)^2}{\sum_j (y_j - \bar{y})^2}, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_j (y_j - \hat{y}_j)^2} \tag{12}$$

pooled over all points of a test set, and per mix. All computations ran on a single CPU core.

5 Results

5.1 Numerical verification

The direct solver behaves as expected (Appendix A, Table A.1). For a constant diffusivity it reproduces the analytical series solution of slab drying to within 2 × 10⁻³. Under mesh refinement the slab solution converges at second order, with an error of 9 × 10^{-5} in S_{avg} at 80 cells, and the water lost through the drying face balances the change in stored water to machine precision (relative error 10^{-15}). The disc solution on the 24 × 16 mesh differs from a 96 × 64 reference by at most 4 × 10^{-3}, mostly in the first hours of drying, before the first measurement. Refitting eight mixes on a 36 × 24 mesh changed the identified parameters by at most 0.001 decade for D_v and 0.007 decade for K_{int}, well inside their uncertainty

(Section 5.2). The tabulated solver differs from the direct solver by at most 9×10^{-5} in S_{avg}, and classical fits obtained with it agree with direct fits to within 0.0003 decade for D_v and 0.002 decade for K_{int}. Differences between the learning methods compared below are therefore not caused by numerical approximation.

5.2 Identifiability of the transport parameters

With α_0 and n fixed, the disc model reproduces every drying curve with R^2 between 0.9917 and 0.9995 (median 0.9986, median RMSE 0.0060). Freeing more parameters does not help (Table 3). When D_v, α_0 and K_{int} are fitted together, D_v and α_0 are perfectly correlated ($|\rho|$ = 1.00) and the condition number of the Jacobian rises to 4×10^2–2×10^3; adding n leaves parameter pairs correlated above 0.99. With two free parameters the correlation is 0.72–0.82 and the condition number 12–16. The total misfit decreases with n and levels off beyond $n \approx 8$: on the five training mixes of the profile it is 49% higher at $n = 2$ and only 2% lower at $n = 16$, the same pattern as with the withheld mix included. The value $n = 8$ lies in the range reported by Bažant and Najjar [9]. Because D_v and α_0 cannot be separated, the identified D_v is conditional on $\alpha_0 = 0.05$.

Table 3. Jacobian-based identifiability with the disc model ($n = 8$ when not identified): fit quality, largest parameter correlation and condition number.

Mix	Identified	R^2	RMSE	max $\lvert\rho\rvert$	cond(**J**)
C-0.4	$D_v, \alpha_0, K_{\mathrm{int}}$	0.9974	0.0074	1.00	1.3×10^3
C-0.4	$D_v, \alpha_0, n, K_{\mathrm{int}}$	0.9989	0.0048	1.00	1.1×10^4
C-0.4	D_v, K_{int}	0.9975	0.0073	0.77	14
F20-0.3	$D_v, \alpha_0, K_{\mathrm{int}}$	0.9985	0.0061	1.00	1.5×10^3
F20-0.3	$D_v, \alpha_0, n, K_{\mathrm{int}}$	0.9985	0.0061	1.00	1.5×10^6
F20-0.3	D_v, K_{int}	0.9984	0.0063	0.82	12
S40-0.5	$D_v, \alpha_0, K_{\mathrm{int}}$	0.9988	0.0053	1.00	1.2×10^3
S40-0.5	$D_v, \alpha_0, n, K_{\mathrm{int}}$	0.9994	0.0036	0.99	2.0×10^4
S40-0.5	D_v, K_{int}	0.9988	0.0052	0.82	16
F20S20-0.5	$D_v, \alpha_0, K_{\mathrm{int}}$	0.9993	0.0038	1.00	4.4×10^2
F20S20-0.5	$D_v, \alpha_0, n, K_{\mathrm{int}}$	0.9993	0.0038	1.00	4.3×10^3
F20S20-0.5	D_v, K_{int}	0.9993	0.0038	0.72	12

The Jacobian gives only a local picture. The profile likelihoods (Fig. 4a–c) show a well-defined minimum for both parameters in every mix, with all 95% intervals inside the admissible range. The residuals of the digitised curves are strongly autocorrelated (median lag-1 correlation 0.63, so that a median of 46 points carries the information of about 11 independent ones). With this correction, D_v is determined to within 0.03 decade (median width of the 95% interval; at most 0.23) and K_{int} to within 0.32 decade, but up to 1.5 decade for F20S20-0.3. The intervals that ignore autocorrelation are about half as wide (median 0.12 decade for K_{int}) and would overstate the precision. The reference diffusivity is thus a precise quantity, whereas the permeability is known only to a few tenths of a decade from a drying curve.

The synthetic tests confirm this asymmetry (Fig. 4d). Inverting exact curves generated on a finer mesh recovers the true parameters to within 0.001 decade for D_v and 0.008 decade for K_{int}, also when only the first 14 days are available (0.007 decade), so the model is structurally identifiable over the experimental time range. Measurement-like perturbations affect mainly K_{int}: independent noise of 0.01 produces errors of up to 0.25 decade (against 0.018 for D_v), correlated noise up to 0.26, and an error of 0.25 d in the time origin up to 0.39 decade. K_{int} controls the capillary stage of the first day, which is described by only a few early points of each curve.

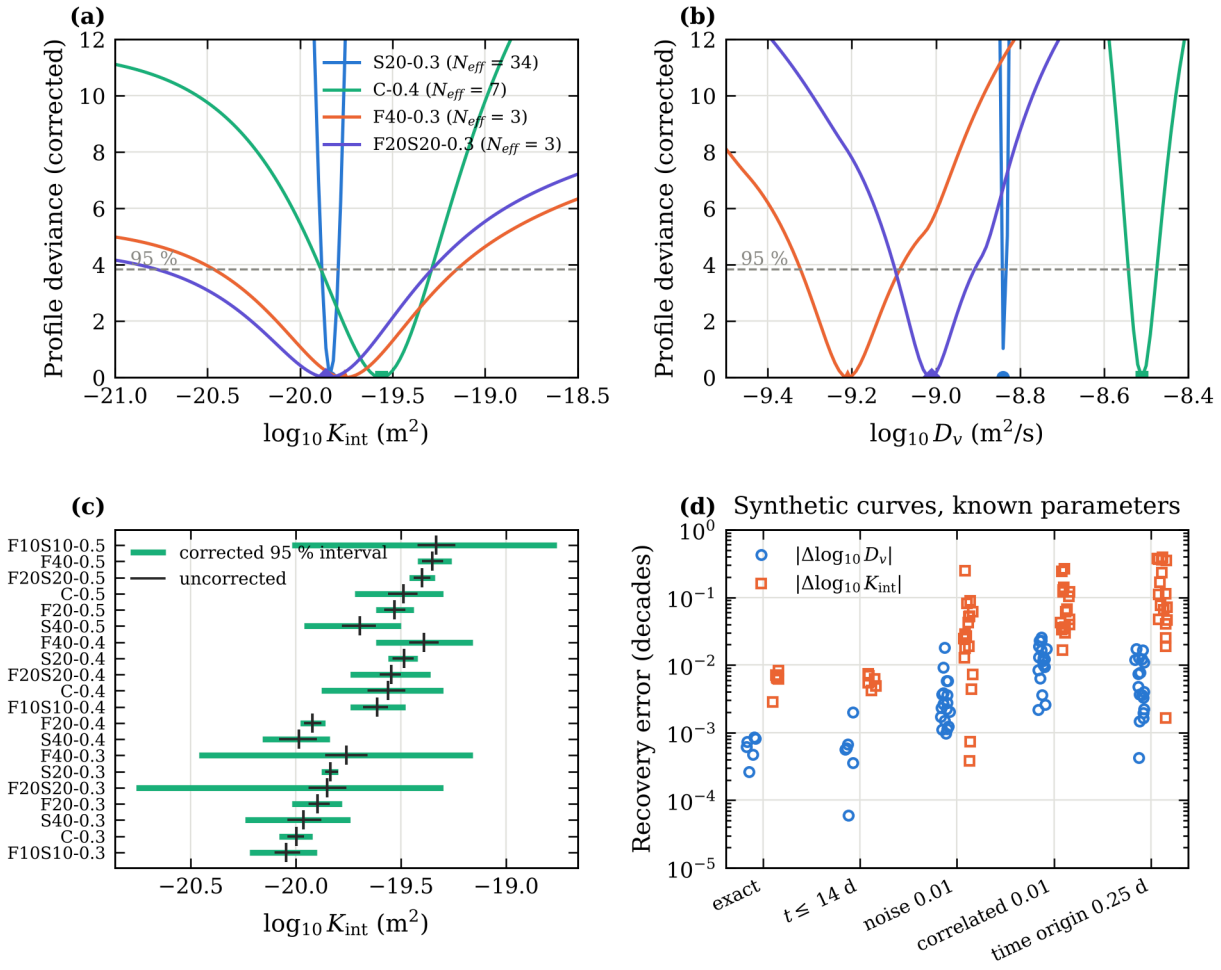


Fig. 4. Identifiability. (a, b) Profile deviance, corrected for residual autocorrelation, of $\log_{10} K_{\mathrm{int}}$ and $\log_{10} D_v$ for four mixes; the dashed line is the 95% threshold. (c) 95% profile intervals of $\log_{10} K_{\mathrm{int}}$ for all mixes, with and without the correction. (d) Recovery errors on synthetic curves with known parameters under five conditions (six parameter sets, three replicates for the noisy conditions).

5.3 Prediction of withheld mixes

Table 4 lists the accuracy on the six withheld mixes and Fig. 5 shows the curves. All methods that use the transport model are close: the PENN reaches R^2 = 0.9931 (RMSE = 0.0137 ± 0.0002 over five seeds), two-stage inversion with linear regression 0.9934 (RMSE 0.0135), and the other two-stage and PENN variants lie between RMSE 0.0130 and 0.0141. The data-driven network is less accurate (RMSE 0.0202), and so is the Gaussian-process regression (RMSE 0.0204), which extrapolates poorly from 14 mixes.

Fitting the model directly to the withheld curves, which uses their data, gives RMSE 0.0063; the difference with the predictive methods measures what is lost when the parameters must be inferred from three composition descriptors.

Table 4. Accuracy on the six withheld mixes (278 points). Networks: mean ± standard deviation over five seeds. [a] After a one-off tabulation of the disc solutions (14 min). [b] Sum of the per-mix fits of the training mixes.

Method	R^2	RMSE	Training time
Two-stage, linear (OLS)	0.9934	0.0135	555 s[b]
Two-stage, ridge	0.9933	0.0136	555 s[b]
Two-stage, quadratic	0.9939	0.0130	555 s[b]
Two-stage, Gaussian process	0.9849	0.0204	555 s[b]
PENN	0.9931 ± 0.0002	0.0137 ± 0.0002	1 s[a]
PENN, revised (linear map, L-BFGS)	0.9937 ± 0.0000	0.0132 ± 0.0000	< 1 s[a]
PENN, linear composition map	0.9928 ± 0.0001	0.0141 ± 0.0001	1 s[a]
PENN, 1 × 8 units	0.9932 ± 0.0000	0.0137 ± 0.0000	1 s[a]
Data-driven network	0.9852 ± 0.0012	0.0202 ± 0.0008	12 s
Reference: classical fit to each withheld mix	*0.9986*	*0.0063*	*41 s per mix*

Table 5. Coefficient of determination for each withheld mix (networks: mean over five seeds).

Mix	Two-stage, linear	PENN	PENN, revised (linear map, L-BFGS)	Two-stage, Gaussian process	Data-driven network
C-0.4	0.994	0.992	0.994	0.997	0.995
C-0.5	0.986	0.999	0.992	0.970	0.897
F10S10-0.3	0.998	0.996	0.998	0.974	0.994
F20-0.3	0.994	0.994	0.995	0.985	0.994
F20S20-0.4	0.998	0.998	0.998	0.993	0.996
S40-0.4	0.981	0.977	0.979	0.976	0.987

The largest differences between methods occur for C-0.5, the most permeable withheld mix and the only plain concrete at w/b = 0.5 (Table 5). The data-driven network underestimates its saturation during the first week (R^2 = 0.897), whereas linear two-stage inversion and the PENN variants predict it with $R^2 \geq 0.98$. In these models, faster drying of a more permeable mix follows from larger parameters, not from a curve shape that must have been seen in training.

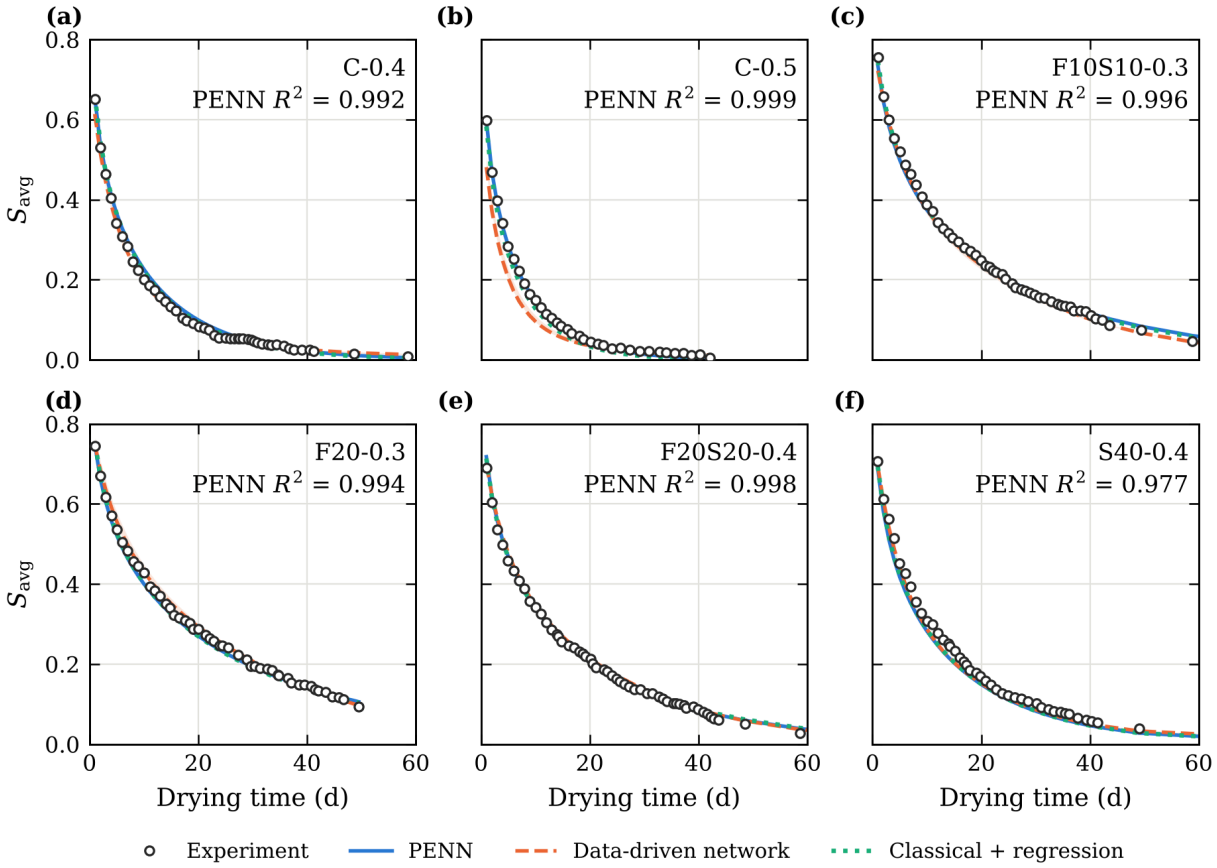


Fig. 5. Drying curves of the six withheld mixes: experiment and predictions. Bands show the range over random seeds; R^2 refers to the PENN.

5.4 Generalisation across mixes and binder families

Cross-validation over all 20 mixes gives the same ranking (Table 6, Fig. 6). In leave-one-mix-out validation, linear and ridge two-stage inversion and all PENN variants reach a pooled R^2 of 0.991–0.992, and the paired per-mix comparison finds no significant difference between the PENN and linear two-stage inversion (mean RMSE difference +0.0006, 95% interval −0.0012 to +0.0020; Wilcoxon p = 0.09). The Gaussian-process and quadratic regressions are significantly worse (p = 0.002 and 0.01).

Withholding a whole binder family separates the models more clearly. Linear two-stage inversion, ridge regression and the PENN variants keep $R^2 \geq 0.962$ for every mix, and again do not differ significantly from one another (PENN against linear inversion: +0.0005, interval −0.0011 to +0.0018, p = 0.22). The flexible models fail for several mixes: the lowest per-mix R^2 is 0.381 for the Gaussian process, 0.198 for quadratic regression and 0.218 for the data-driven network, which has 7 mixes below R^2 = 0.95. The PENN is significantly more accurate than the data-driven network (mean per-mix RMSE difference −0.0172, interval −0.0283 to −0.0078, p = 0.001). The quadratic and Gaussian-process regressions also use the transport model, so the failures are not caused by the absence of physics: they come from composition maps that are too flexible to extrapolate to a binder they have not seen. What generalises is the combination of the transport model with a low-capacity, nearly linear composition map, whether it is fitted in two stages or end to end.

Table 6. Cross-validation over the 20 mixes (one seed per fold). Lowest R^2 and number of mixes below 0.95 refer to per-mix values.

Method	LOMO R^2	LOMO RMSE	LOMO lowest R^2	LOFO R^2	LOFO RMSE	LOFO lowest R^2	LOFO mixes < 0.95
Two-stage, linear (OLS)	0.9919	0.0147	0.974	0.9913	0.0152	0.965	0
Two-stage, ridge	0.9918	0.0148	0.972	0.9916	0.0150	0.966	0
Two-stage, quadratic	0.9870	0.0186	0.965	0.9012	0.0513	0.198	6
Two-stage, Gaussian process	0.9814	0.0223	0.918	0.9484	0.0371	0.381	5
PENN	0.9909	0.0156	0.975	0.9903	0.0160	0.968	0
PENN, revised (linear map, L-BFGS)	0.9916	0.0150	0.974	0.9908	0.0156	0.962	0
PENN, linear composition map	0.9912	0.0153	0.970	0.9899	0.0164	0.969	0
PENN, 1 × 8 units	0.9911	0.0154	0.974	0.9905	0.0159	0.966	0
Data-driven network	0.9868	0.0188	0.953	0.9355	0.0415	0.218	7

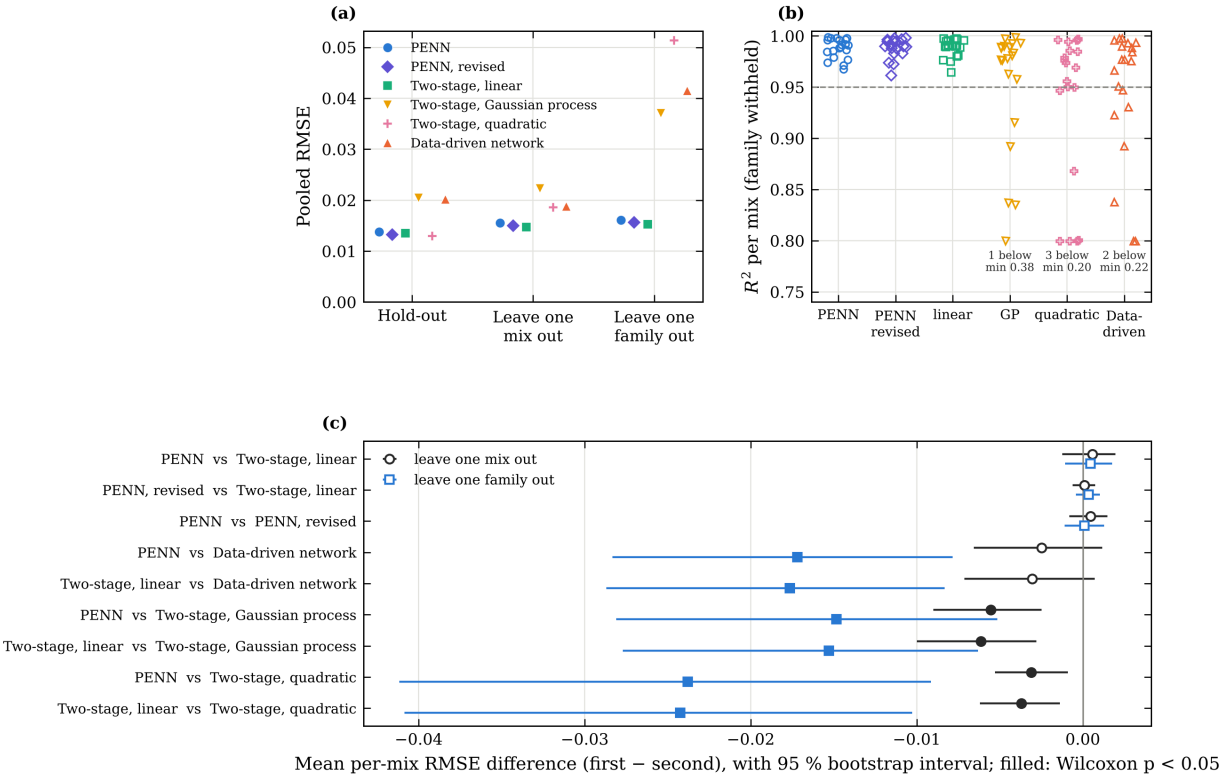


Fig. 6. Model comparison. (a) Pooled RMSE in the three validation schemes. (b) Per-mix R^2 when the whole binder family is withheld; values below 0.80 are drawn at the lower edge and counted. (c) Mean per-mix RMSE differences between pairs of methods with 95% bootstrap intervals over mixes; filled symbols: Wilcoxon $p < 0.05$.

5.5 Robustness and data efficiency

Measurement and processing uncertainty. Perturbing the training data changes the accuracy on the withheld mixes very little for the physics-based models (Fig. 7a, Table 7). With correlated, digitisation-like errors the RMSE becomes 0.0142 for the PENN and 0.0147 for linear two-stage inversion, against 0.0271 for the data-driven network; independent noise and time-origin errors have smaller effects. The reconstruction of the curves is not a sensitive choice: moving the time offset to either end of its

interquartile range, or estimating it from the training mixes only, changes the classical parameters by at most 0.002 decade for D_v and 0.009 decade for K_{int}. Using only the short (≤ 14 d) or only the long series gives the same accuracy. The parameters, however, react differently. The per-mix classical estimates of K_{int} move by up to 0.219 decade under correlated noise and 0.207 decade under time-origin errors (Fig. 7b), consistent with the profile intervals, whereas parameters predicted from composition move less, because they pool all training curves: by up to 0.133 decade for the revised PENN. The original PENN moves by only 0.009 decade, but this stability partly reflects the compression of K_{int} discussed in Section 5.6.

Table 7. Robustness: RMSE on the unperturbed withheld curves after perturbing the training data, and largest shift of $\log_{10} K_{\mathrm{int}}$ over the 20 mixes (classical: per-mix fits; PENN: predicted parameters). Noise variants: mean of five replicates.

Perturbation of the training data	RMSE linear	RMSE PENN	RMSE PENN rev.	RMSE data-driven	max Δlog K classical	max Δlog K PENN	max Δlog K PENN rev.
None (reference)	0.0135	0.0137	0.0132	0.0202	–	–	–
Independent noise, s.d. 0.01	0.0137	0.0139	0.0134	0.0205	0.070	0.014	0.032
Independent noise, s.d. 0.02	0.0139	0.0140	0.0136	0.0214	0.145	0.032	0.069
Correlated noise, s.d. 0.01	0.0147	0.0142	0.0142	0.0271	0.219	0.009	0.133
Time-origin error, s.d. 0.25 d	0.0140	0.0140	0.0137	0.0203	0.207	0.033	0.060
Offset 0.97 d	0.0133	0.0136	0.0131	0.0201	0.009	0.005	0.010
Offset 1.13 d	0.0136	0.0138	0.0133	0.0203	0.009	0.005	0.010
Offset from training mixes	0.0135	0.0138	0.0132	0.0202	0.001	0.001	0.002
Short series only	0.0148	0.0135	0.0135	0.0245	0.197	0.090	0.223
Long series only	0.0136	0.0138	0.0133	0.0194	0.170	0.007	0.062

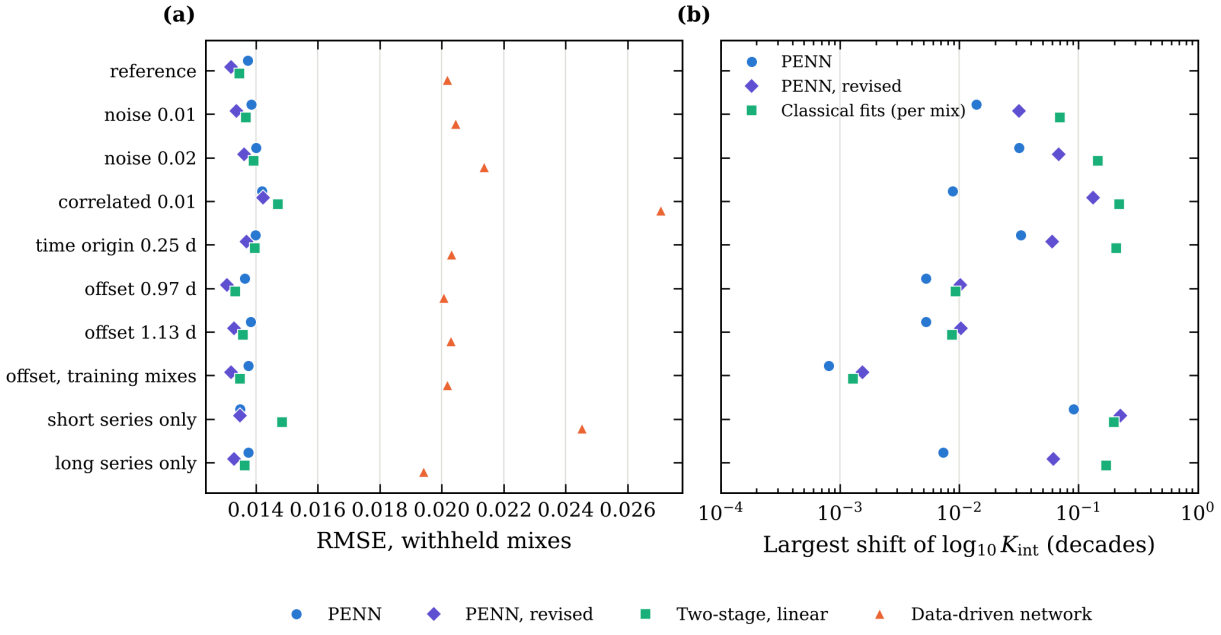


Fig. 7. Robustness. (a) RMSE on the withheld mixes after perturbing the training data. (b) Largest shift of the identified or predicted $\log_{10} K_{\mathrm{int}}$ relative to the unperturbed reference.

Number of training mixes. With 8, 11 or 14 training mixes, linear two-stage inversion and the PENN variants are equivalent; with 14 mixes the median RMSE over ten random draws is 0.0142 for linear inversion, 0.0162 for the PENN and 0.0159 for the revised PENN (Fig. 8a,b). With only five mixes, the original PENN is more accurate and more stable (median RMSE 0.0220, worst-mix R^2 0.92) than linear inversion (0.0256, 0.89) and ridge regression (0.0268), whose four regression coefficients are poorly constrained by five points; the revised PENN, an unregularised linear map, behaves like linear inversion (0.0275). The advantage therefore comes from the regularisation of the original network rather than from end-to-end training as such. The data-driven network and the Gaussian process are the least accurate at every training size.

Short drying curves. The clearest advantage of end-to-end training appears when the training curves are short (Fig. 8c). With curves cut after 7 days (84 points instead of 654), the PENN predicts the complete curves of the withheld mixes with RMSE 0.0138, against 0.0165 for two-stage inversion and 0.0314 for the data-driven network. With curves cut after 3 or 4 days, two-stage inversion can only use the 11 mixes that still have at least three points, and its error rises to 0.0258; the PENN, which uses every point of every training mix, stays at 0.0180. The revised PENN, whose linear map is not regularised, lies in between (RMSE 0.0222 at 4 days and 0.0143 at 7 days). Per-mix inversion of a short curve is poorly conditioned, whereas the PENN constrains the parameters of all mixes jointly through the composition map; with very little data, the regularisation of the original PENN is an advantage rather than a bias.

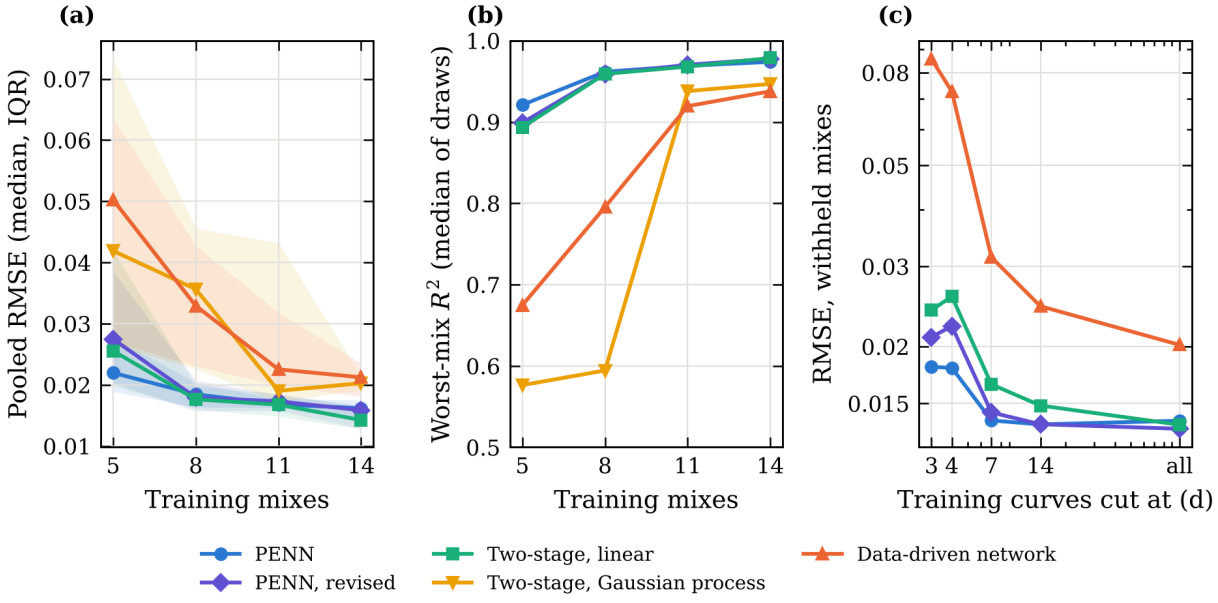


Fig. 8. Data efficiency. (a) Median pooled RMSE (band: interquartile range over ten random draws of the training mixes) and (b) median worst-mix R^2 against the number of training mixes. (c) RMSE on the complete withheld curves when the training curves are cut after a given drying time.

5.6 End-to-end learning and parameter recovery

Accuracy on withheld curves does not show whether the learned parameters are right. The composition maps were therefore also tested on synthetic curves generated, for the 20 mixes and at their measurement times, from known parameters that depend linearly on composition, with noise of 0.01 added. Linear two-stage inversion recovers the permeability of the withheld mixes with a mean error of 0.024 decade. The PENN in its original setting recovers D_v well but compresses the variation of K_{int} between mixes: the

slope of estimated against true $\log_{10} K_{\mathrm{int}}$ over the training mixes is only 0.17, and the error on the withheld mixes is 0.088 decade. Training longer with the same weight decay does not help (slope 0.16). Removing the weight decay restores the slope (0.86) on synthetic data, but on the experimental data the unregularised 2 × 16 network then over-fits the composition map: its hold-out RMSE rises to 0.0187 and its lowest per-mix R^2 in leave-one-family-out validation falls to 0.78. Weight decay is thus what lets the original PENN generalise, and also what biases its permeabilities.

A linear composition map cannot over-fit in this way. Trained with Adam and a decaying step, it remains under-converged in the weakly sensitive K_{int} direction (slope 0.85); trained to convergence with L-BFGS and without weight decay, it recovers K_{int} as well as two-stage inversion (slope 0.96, mean error 0.022 decade on the withheld mixes). When the synthetic truth also contains mix-to-mix scatter not explained by composition, the two have similar errors on the withheld mixes (0.130 and 0.139 decade). This revised PENN was selected on synthetic data only and then evaluated once on the experimental data. Its predictive accuracy is close to that of the original PENN and of linear two-stage inversion (hold-out RMSE 0.0132; lowest leave-one-family-out R^2 0.962; Tables 4 and 6), and its permeabilities follow the classical estimates more closely (correlation 0.78, against 0.48 for the original PENN), falling inside the 95% profile interval of the classical estimate for 13 of the 20 mixes (original PENN: 9). A hidden regulariser or an under-converged optimiser can therefore bias a weakly identifiable parameter without any visible effect on the predicted curves; only the synthetic recovery test revealed it.

5.7 PENN versus PINN consistency

The PINN was compared with the PENN on the slab model with identical constants, data split and parameter network (Table 8). On the withheld mixes the PINN is less accurate (RMSE 0.0216 against 0.0133) and varies more between seeds, although it fits the training data far more closely (training RMSE 0.0033 against 0.0125). Without its PDE residual it degrades further (RMSE 0.0365), so the residual does carry information.

The solver check is decisive (Fig. 9a,b). Run with the PENN parameters, the direct solver reproduces the PENN output with R^2 = 0.9998. Run with the PINN parameters it does not (R^2 = 0.860): the PINN reference diffusivities are close to the classical values (mean difference 0.06 decade), but its permeabilities are about 1.1 decade too low.

Four diagnostic runs locate the cause (Fig. 9c,d and Appendix D). Starting the PINN from the classical parameters does not prevent the drift (bias −1.08 decade), and a ten times larger PDE weight barely changes it (−1.03). Training on synthetic curves generated by the slab solver from the classical parameters, i.e. without model error or noise, gives the same bias (−1.13) although the PINN then fits the data to RMSE 0.0003. Finally, with the parameters fixed at their classical values, the PINN reaches the same total loss as with free parameters (0.135 against 0.135), with a lower data misfit but a larger PDE residual (RMS 0.107 against 0.048), and its parameters then pass the solver check (R^2 = 0.9988). The PINN loss is therefore nearly flat along K_{int}: the field network finds it easier to satisfy the residual with a too-low permeability, which slows the early capillary stage that the similarity ansatz represents least well, and the data term cannot tell the difference because that stage ends before the first measurement. The inconsistency is a property of the soft-constrained formulation for this problem, not of the data, the initialisation or the weighting. The saturation profiles of the two models differ most during the first two days (Fig. D.2).

Table 8. Controlled comparison on the one-dimensional slab model (six withheld mixes). Solver check: R^2 between each model and the direct solver run with the parameters returned by that model.

Method	R^2 (withheld)	RMSE (withheld)	RMSE (training)	Solver check R^2
PENN (differentiable solver)	0.9935 ± 0.0001	0.0133 ± 0.0001	0.0125	0.9998
Classical + regression	0.9938	0.0131	–	exact
PINN	0.9826 ± 0.0071	0.0216 ± 0.0043	0.0033	0.860
PINN without PDE residual	0.9411 ± 0.0640	0.0365 ± 0.0210	0.0031	–
Data-driven network	0.9852 ± 0.0012	0.0202 ± 0.0008	0.0059	–

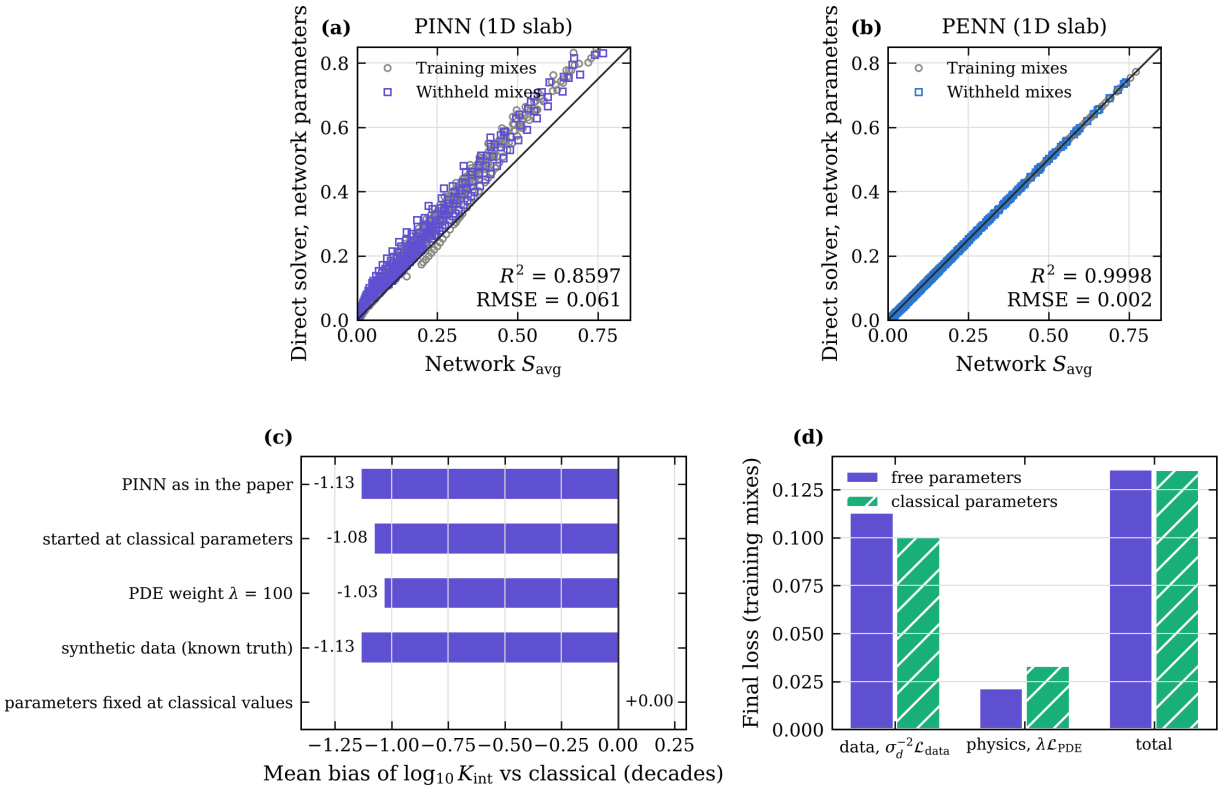


Fig. 9. PINN consistency on the slab model. (a, b) Average saturation computed by the direct solver with the parameters returned by the PINN and the PENN, against the output of each network. (c) Mean bias of the PINN permeability relative to the classical estimates in the diagnostic runs; a PINN with parameters fixed at the classical values has none. (d) Final loss terms with free and with classical parameters.

5.8 Transport parameters and their physical interpretation

Figure 10 compares the classical estimates, with their 95% profile intervals, with the parameters predicted from composition alone. The reference diffusivities agree closely (mean absolute difference 0.025 decade for the PENN, 0.023 for the revised PENN). Both parameters increase with w/b. For plain concrete, D_v rises from $10^{-8.76}$ to $10^{-8.36}$ m²/s between w/b = 0.3 and 0.5, consistent with the coarser and better-connected capillary porosity of high-w/b pastes [1, 3]. At a given w/b all SCMs lower D_v. The effect is largest for 40% fly ash, whose reference diffusivity at w/b = 0.3 ($10^{-9.15}$ m²/s) is 2.5 times lower than that of plain concrete, followed by the 20% + 20% ternary blend and 20% fly ash; slag reduces D_v less, and 20% slag least. This ranking agrees with the pore refinement produced by the pozzolanic and latent

hydraulic reactions [1]. The permeabilities increase with w/b as well, but their profile intervals often overlap between neighbouring mixes, so differences of less than about 0.3 decade between individual mixes should not be interpreted.

The diffusivity varies by about an order of magnitude with saturation (Fig. 11), between 4×10^{-11} and 2×10^{-10} m²/s in the dry, vapour-dominated range and up to 7×10^{-10}–2×10^{-9} m²/s near $S = 0.79$, where capillary transport takes over. The position of this transition is set by S_{cap} and is not identified from the data.

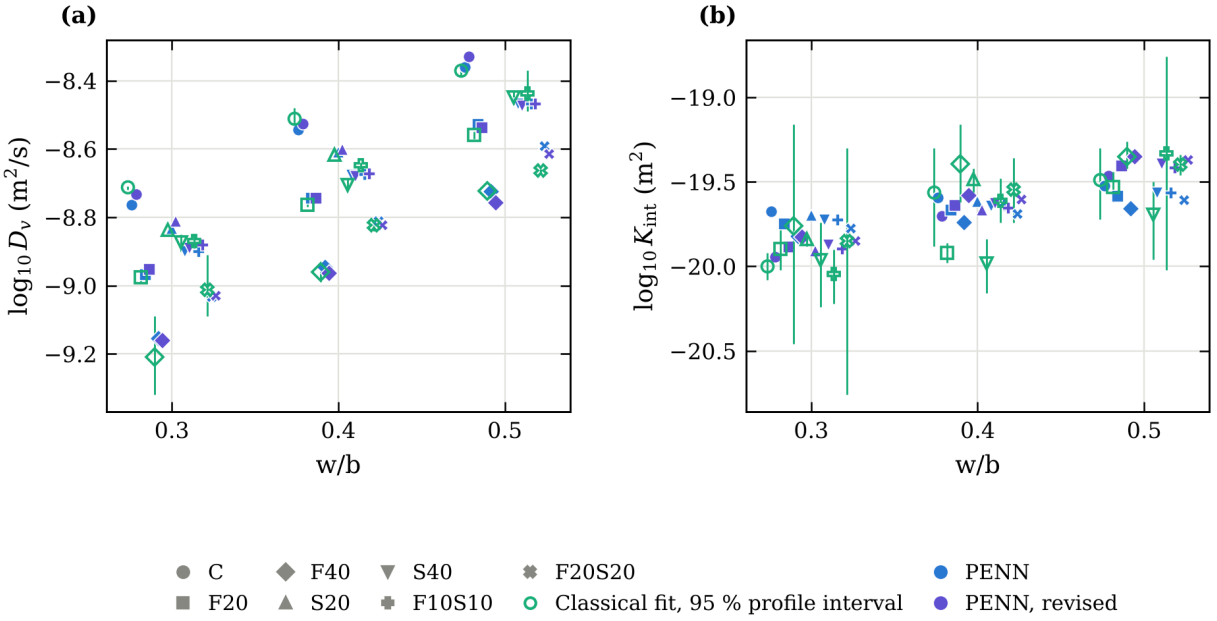


Fig. 10. Transport parameters against w/b: (a) reference diffusivity and (b) intrinsic permeability. Open symbols: classical estimates with 95% profile intervals (corrected for autocorrelation). Filled symbols: predictions from composition by the PENN and the revised PENN (hold-out models).

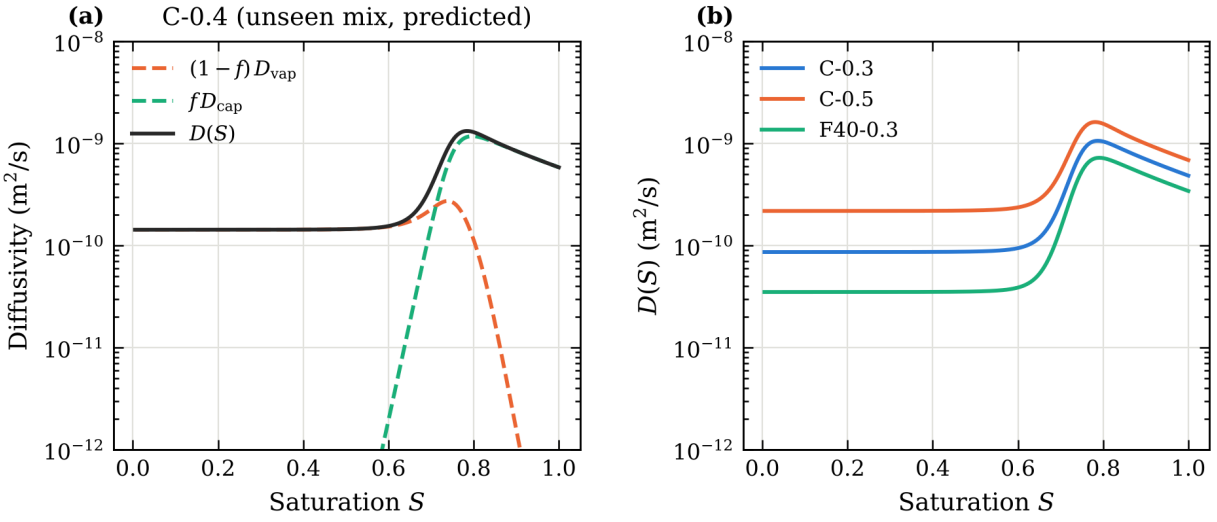


Fig. 11. Diffusivity predicted by the PENN: (a) vapour and capillary contributions for the withheld mix C-0.4; (b) effect of w/b and fly ash.

5.9 Sensitivity to the modelling assumptions

The identified parameters are conditional on the fixed constants of Table 2. Changing one constant at a time within plausible bounds and refitting eight representative mixes with the direct solver shows how strong this dependence is (Fig. 12, Table 9). The minimum-to-maximum ratio α_0 acts almost only on D_v, which shifts by +0.39 decade for α_0 = 0.02 and −0.30 decade for α_0 = 0.10, as expected from the perfect correlation of the two parameters (Section 5.2). The constants that shape the transition between the vapour and capillary regimes (S_c, S_{cap}, k, β and n) act mainly on K_{int}, by up to 0.4 decade for an individual mix. None of these changes alters the fit quality by more than 6%, so the data cannot arbitrate between them. The shifts are, however, nearly uniform across mixes: the rank correlation of the parameters with the reference estimates is at least 0.88. Comparisons between mixes, and the composition trends of Section 5.8, are therefore robust, whereas absolute values carry an uncertainty of a few tenths of a decade from the constitutive assumptions alone, in addition to the statistical uncertainty of Section 5.2.

Two boundary assumptions were also tested. A finite surface exchange (surface time constant 0.1 d) changes K_{int} by at most 0.08 decade, and an initial saturation of 0.97 instead of 1 by at most 0.22 decade. A sealed lateral surface, which reduces the problem to one-dimensional drying through the two flat faces, fits the data clearly worse (RMSE 2.18 times larger), shifts K_{int} by up to 1.16 decade and drives two fits to the bound of the admissible range. This supports the assumption that the discs dried on all faces.

Table 9. Sensitivity of the classical estimates to the modelling assumptions (eight mixes): mean shift of the parameters, ratio of the fit RMSE to the reference, and Spearman rank correlation with the reference estimates.

Change (reference value)	**mean Δlog D_v**	**mean Δlog K_{int}**	**max | Δlog K_{int} |**	**RMSE ratio**	**rank corr. D_v**	**rank corr. K_{int}**
Finer mesh (36 × 24)	−0.001	+0.007	0.007	1.00	1.00	1.00
Lateral surface sealed (1D, L = 25 mm)	+0.296	+0.938	1.161	2.18	1.00	0.83
α_0 = 0.02 (0.05)	+0.393	−0.121	0.193	1.01	1.00	0.98
α_0 = 0.10 (0.05)	−0.299	+0.034	0.051	1.00	1.00	1.00
n = 4 (8)	−0.050	−0.248	0.377	1.06	1.00	0.88
n = 16 (8)	+0.003	+0.047	0.067	1.00	1.00	1.00
S_c = 0.75 (0.80)	−0.011	−0.246	0.411	1.02	1.00	0.93
S_{cap} = 0.70 (0.75)	−0.008	−0.172	0.246	1.03	1.00	1.00
S_{cap} = 0.80 (0.75)	+0.005	+0.114	0.284	0.99	1.00	0.93
k = 25 (50)	−0.012	−0.125	0.149	1.03	1.00	1.00
β = 2.0 (2.5)	+0.000	+0.142	0.145	1.00	1.00	1.00
β = 3.0 (2.5)	−0.000	−0.125	0.128	1.00	1.00	1.00
S_{init} = 0.97 (1)	+0.005	−0.141	0.220	1.00	1.00	1.00
Surface exchange, τ_b = 0.1 d (10^{-4} d)	−0.002	+0.048	0.084	1.02	1.00	1.00
P_{c0} or η_l × c	0	$-\log_{10}$ c (exact)	–	1.00	1.00	1.00

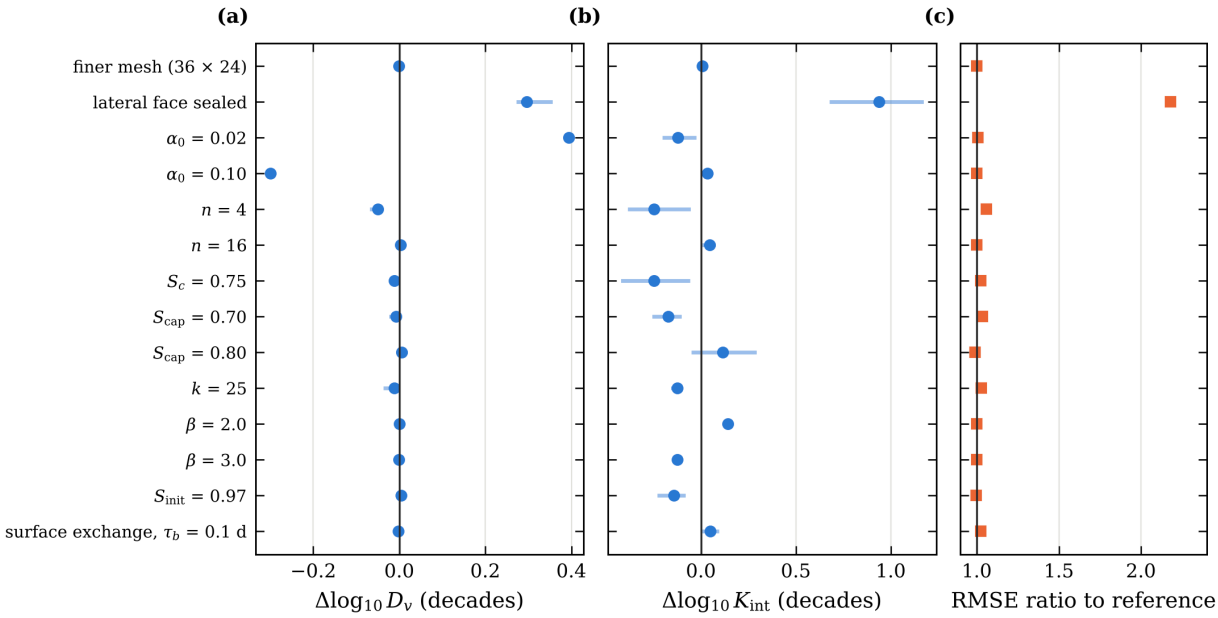


Fig. 12. Sensitivity of the identified parameters to the modelling assumptions: mean (dot) and range (bar) over eight mixes of the shift of (a) $\log_{10} D_v$ and (b) $\log_{10} K_{int}$, and (c) ratio of the fit RMSE to the reference model.

5.10 Internal saturation fields

Because the PENN runs the transport model, every prediction comes with the saturation field inside the specimen (Fig. 13). After one day a dry layer a few millimetres thick has formed along the exposed faces, while the centre is still almost saturated (S = 0.97 for C-0.4). The centre falls below S = 0.5 after about 14 days for C-0.4, 18 days for S40-0.4 and 35 days for F20-0.3. Such fields are what durability models need as initial conditions for chloride ingress or carbonation analyses. They are model predictions, conditional on the assumptions of Section 3, and have not been compared with measured profiles.

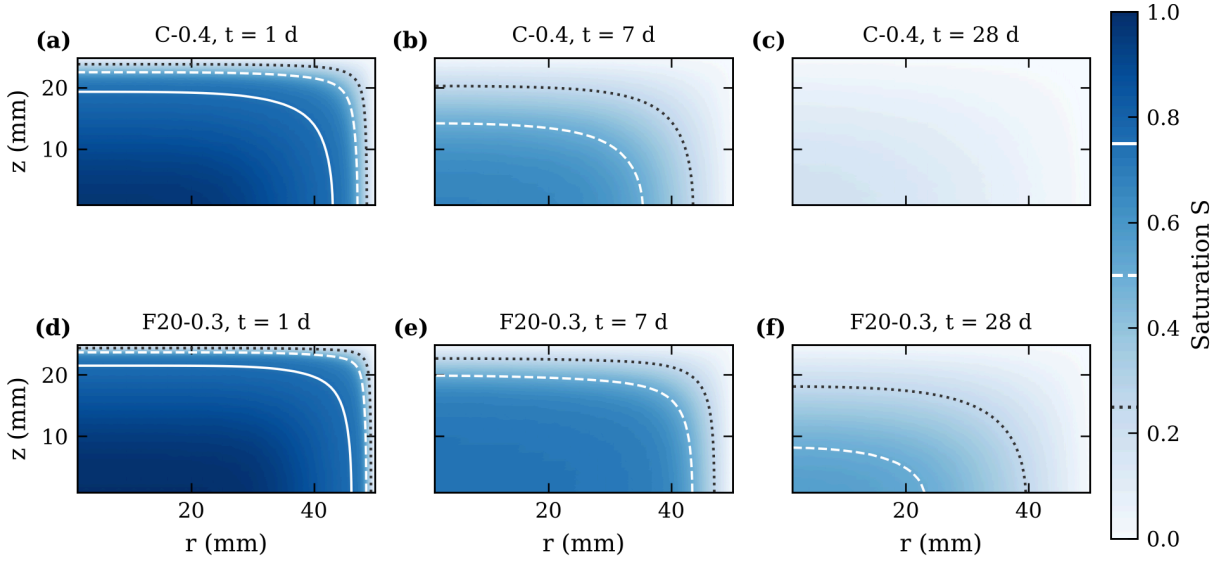


Fig. 13. Saturation in one quarter of the disc (drying faces at the top and right) predicted by the PENN for the withheld mixes C-0.4 and F20-0.3 after 1, 7 and 28 days. Contours: S = 0.75 (solid), 0.50 (dashed) and 0.25 (dotted).

6 Discussion

6.1 What the physics-embedded network adds, and what it does not

The results do not support the claim that a physics-embedded network predicts drying more accurately than classical identification. On complete curves, with 8 or more training mixes, the PENN and two-stage inversion with a linear composition map are statistically indistinguishable, and the simpler method is slightly ahead on average. What makes both of them generalise is the same: a verified transport model, which turns composition into a full drying curve through two physically meaningful parameters, and a composition map simple enough to extrapolate. Flexible maps fail when a binder family is withheld whether or not the transport model is used, as the Gaussian-process and quadratic regressions show.

End-to-end training is useful in a narrower but practically relevant situation: when the individual curves do not constrain their parameters well. With curves of a few days, the PENN pools the information of all curves through the composition map and remains accurate where per-mix inversion becomes unstable; with very few training mixes, the regularised original PENN is the most stable model, an effect of its regularisation rather than of end-to-end training itself. It is also convenient: a new mix requires one network evaluation, curves of different length are used as they are, and further descriptors can be added without changing the solver. The price is a tabulated or differentiable solver and, as Section 5.6 showed, the need to check that the training settings do not bias the parameters. We would recommend synthetic recovery tests as a routine part of any such model.

6.2 Accuracy is not identification

Three observations in this study separate fitting curves from identifying parameters. First, the classical fits reach $R^2 > 0.99$ for every mix, yet the profile likelihood shows that K_{int} is uncertain by a few tenths of a decade, up to 1.5 decade, once the autocorrelation of digitised residuals is acknowledged. Second, the original PENN and the revised PENN predict the withheld curves equally well but differ in how faithfully they reproduce the variation of K_{int} between mixes. Third, the PINN fits the training curves best of all methods and returns the least consistent permeabilities. In each case the predicted average saturation hides an error in a parameter that matters for other uses of the model: K_{int} governs the capillary regime, which dominates wetting and therefore chloride ingress under cyclic exposure. Parameters intended for durability modelling should be reported with profile intervals and checked by running the identified model in an independent solver.

The PINN result should not be generalised to all PINNs. It holds for this formulation, with hard-constrained boundary and initial conditions and a similarity ansatz, on a stiff nonlinear diffusion problem observed through spatial averages. Our diagnostics show that it is not a matter of initialisation or loss weighting, and the loss-landscape difficulties reported for PINNs [25, 28] are consistent with it. Other architectures or training strategies may do better, but any PINN used for parameter identification should be checked against an independent solver.

6.3 Limitations

The study rests on one published dataset. The curves were digitised from figures, their curing age and the sealing condition of the lateral surface are not recorded in the data file, and eight mixes (w/b = 0.6 and S20-0.5) could not be used. The digitisation errors are autocorrelated and affect K_{int} in particular. The transport model is isothermal, at the 50 °C of the oven, and uses fixed constitutive constants; transferring

the parameters to service temperatures requires a temperature-dependent model. The identified parameters are effective values conditional on these constants and should not be read as intrinsic properties without independent measurements, such as sorption isotherms, permeability tests or measured moisture profiles. No such measurements were available, so the identified parameters have been checked for consistency with the model, not validated experimentally. The absorption tests of the same programme [8] could provide such a validation if the raw data become available. Finally, the internal saturation fields are model predictions.

6.4 Engineering implications

For mix design and durability assessment, the study suggests a simple workflow. A drying test of moderate length gives a precise reference diffusivity and an order-of-magnitude permeability for each tested mix. Across a family of mixes, a transport model with a linear composition map, fitted in two stages or end to end, predicts the drying of untested compositions within the binder systems represented, and extrapolates to a new binder system with an accuracy close to that of a direct fit. Flexible machine-learning models trained on the same small database should not be used for such extrapolation. Where only short tests are available, joint end-to-end training is preferable to per-mix inversion. In all cases, the identified parameters inherit the uncertainty of the fixed constitutive constants, which should be stated when they are transferred to service-life models.

7 Conclusions

A vapour–capillary moisture transport model of the actual disc specimens was combined with several ways of learning its parameters from mix composition, using a reconstructed public database of 20 plain, binary and ternary concretes. The main conclusions are:

1. Average-saturation curves identify two parameters per mix. The reference diffusivity is determined to within 0.03 decade; the intrinsic permeability to within 0.32 decade (median 95% intervals corrected for autocorrelated residuals). Both depend on fixed constitutive constants, which shift them by up to 0.4 decade but preserve the ranking of the mixes.
2. A physics-embedded neural network predicts six withheld mixes with R^2 = 0.9931 (RMSE 0.0137). Classical inversion followed by linear regression on composition is equally accurate (RMSE 0.0135), and the two do not differ significantly in leave-one-mix-out or leave-one-binder-family-out validation.
3. Generalisation to a new binder family requires the transport model and a low-capacity composition map. Gaussian-process, quadratic and data-driven models fail for several mixes (lowest R^2 0.20), whereas linear two-stage inversion and the PENN keep $R^2 \geq 0.962$ for every mix.
4. End-to-end training is more robust than per-mix inversion when the training curves are short (RMSE 0.0138 against 0.0165 with 7-day curves), and a regularised PENN is the most stable model when only a few mixes are available. Weight decay in the network compresses the permeability towards its mean; synthetic recovery tests revealed this, and a linear composition map trained to convergence without weight decay removes the bias without loss of predictive accuracy.
5. A PINN trained on the same data fits the curves but returns permeabilities about 1.1 decade too low, which fail an independent solver check. The bias persists with exact synthetic data, a warm start and a larger PDE weight, and disappears only when the parameters are fixed: it is a property of the soft-constrained formulation for this problem.

Future work should validate the identified parameters against independent measurements, in particular absorption tests and measured moisture profiles, extend the database beyond one laboratory, and propagate the parameter uncertainty to chloride-ingress predictions [11, 38].

Declarations

CRediT authorship contribution statement. Abderrahmane Sghuri: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft. Xuande Chen: Conceptualization, Supervision, Validation, Writing – review & editing, Funding acquisition. Xiaodong Wang: […].

Data and code availability. The experimental data are those of Zhuang et al. [8]. The digitised and reconstructed data and the code that produces every result, table and figure of this paper are available at https://github.com/Abderahmane2002/hybrid-pinn-lstm-concrete [archive the release on Zenodo and cite its DOI].

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding. [Funding agency and grant number]

Declaration of generative AI use. [Elsevier requires authors to disclose the use of generative AI tools in preparing the manuscript or code; state the tool and its purpose here.]

Appendix A. Numerical verification

Table A.1. Verification of the solvers: maximum error in S_{avg} between 0.25 and 90 days for three representative parameter sets, against a fine reference solution (640 cells for the slab solver, a 160-cell BDF solution for the differentiable solver, 96 × 64 cells for the disc).

Check	Setting	Max. error in S_{avg}
1D finite-volume solver (BDF)	20 cells	1.4×10^{-3}
1D finite-volume solver (BDF)	40 cells	3.4×10^{-4}
1D finite-volume solver (BDF)	60 cells	1.5×10^{-4}
1D finite-volume solver (BDF)	80 cells	8.5×10^{-5}
1D finite-volume solver (BDF)	160 cells	2.1×10^{-5}
1D differentiable solver	60 time steps	1.2×10^{-2}
1D differentiable solver	120 time steps	5.9×10^{-3}
1D differentiable solver	240 time steps	3.0×10^{-3}
1D differentiable solver	480 time steps	1.6×10^{-3}
1D differentiable solver	960 time steps	8.1×10^{-4}
Disc model (BDF)	12 × 8 cells	3.8×10^{-2}
Disc model (BDF)	24 × 16 cells	3.7×10^{-3}
Disc model (BDF)	48 × 32 cells	7.8×10^{-4}
Global mass balance, 1D	28 days	1.2×10^{-15} (relative)

Appendix B. Reproducibility of the original results

Before any change was made, the complete original pipeline was re-run from a clean copy of the code, starting from the raw digitised file: curve reconstruction, classical fits, tabulation of the disc solutions, all network trainings, cross-validation and verification. Table B.1 compares the regenerated results with those of the original analysis. 44 of 44 compared quantities agree within the stated tolerances.

Table B.1. Independent re-run of the original pipeline. Tolerances allow for the random seeds of multi-start optimisation and, for the PINN, for non-deterministic floating-point reductions.

Quantity	Original	Re-run	Status
Cleaned data: number of points	1001	1001	reproduced
Cleaned data: largest difference in S	0	0	reproduced
Cleaned data: largest difference in t (d)	0	0	reproduced
Disc: classical $\log_{10} D_v$, largest difference	0	0.0001568	reproduced
Disc: classical $\log_{10} K_{\mathrm{int}}$, largest difference	0	0.001002	reproduced
Disc: classical fits, median R^2	0.9986	0.9986	reproduced
Slab: classical $\log_{10} D_v$, largest difference	0	0	reproduced
Slab: classical $\log_{10} K_{\mathrm{int}}$, largest difference	0	0	reproduced
Slab: classical fits, median R^2	0.9988	0.9988	reproduced
Disc: PENN hold-out R^2	0.9931	0.9931	reproduced
Disc: PENN hold-out RMSE	0.01374	0.01374	reproduced
Disc: data-driven hold-out R^2	0.9852	0.9852	reproduced
Disc: data-driven hold-out RMSE	0.02018	0.02018	reproduced
Disc: two-stage hold-out R^2	0.9934	0.9934	reproduced
Disc: two-stage hold-out RMSE	0.01347	0.01347	reproduced
Disc: direct fit hold-out R^2	0.9986	0.9986	reproduced
Disc: direct fit hold-out RMSE	0.0063	0.0063	reproduced
Disc: solver check PENN R^2	1	1	reproduced
Slab: PENN hold-out R^2	0.9935	0.9935	reproduced
Slab: PENN hold-out RMSE	0.01333	0.01333	reproduced
Slab: data-driven hold-out R^2	0.9852	0.9852	reproduced
Slab: data-driven hold-out RMSE	0.02018	0.02018	reproduced
Slab: two-stage hold-out R^2	0.9938	0.9938	reproduced
Slab: two-stage hold-out RMSE	0.01311	0.01311	reproduced
Slab: PINN hold-out R^2	0.9826	0.9826	reproduced
Slab: PINN hold-out RMSE	0.02159	0.02159	reproduced
Slab: PINN without PDE hold-out R^2	0.9411	0.9411	reproduced
Slab: PINN without PDE hold-out RMSE	0.03646	0.03646	reproduced
Slab: solver check PENN R^2	0.9998	0.9998	reproduced
Slab: solver check PINN R^2	0.8597	0.8597	reproduced
CV LOFO two-stage pooled R^2	0.9913	0.9913	reproduced
CV LOFO two-stage worst-mix R^2	0.9647	0.9647	reproduced
CV LOFO data-driven pooled R^2	0.9355	0.9355	reproduced
CV LOFO data-driven worst-mix R^2	0.2177	0.2177	reproduced
CV LOFO PENN pooled R^2	0.9903	0.9903	reproduced
CV LOFO PENN worst-mix R^2	0.9677	0.9677	reproduced
CV LOMO two-stage pooled R^2	0.9919	0.9919	reproduced
CV LOMO two-stage worst-mix R^2	0.9736	0.9736	reproduced
CV LOMO data-driven pooled R^2	0.9868	0.9868	reproduced

Quantity	Original	Re-run	Status
CV LOMO data-driven worst-mix R^2	0.9533	0.9533	reproduced
CV LOMO PENN pooled R^2	0.9909	0.9909	reproduced
CV LOMO PENN worst-mix R^2	0.9746	0.9746	reproduced
Numerical verification (Table A.1): largest relative difference	0	0.01562	reproduced
Slab-disc geometry check: largest relative difference	0	0.0005191	reproduced

Appendix C. Hyper-parameter check

The settings of the PENN and of the data-driven network were compared with alternatives in an inner leave-one-family-out cross-validation on the 14 training mixes only (Table C.1). For the PENN, the original setting ranks 5 of 9 configurations, with an inner RMSE (0.0178) within 0.001 of the best (0.0173); the configuration selected by the inner validation gives RMSE 0.0138 on the withheld mixes, the same as the original setting. For the data-driven network the inner validation prefers a stronger weight decay (inner RMSE 0.0346 against 0.0826), but that configuration is less accurate on the withheld mixes (RMSE 0.0309) than the original one. The comparisons in the main text therefore do not favour the PENN through its settings.

Table C.1. Inner cross-validation (leave one family out within the training mixes): RMSE of every configuration tested.

Model	Configuration	Inner RMSE
PENN	2 × 8 tanh, 200 iterations	0.0180
PENN	2 × 8 tanh, 400 iterations	0.0179
PENN	2 × 8 tanh, 800 iterations	0.0179
PENN	2 × 16 tanh, 200 iterations	0.0179
PENN	2 × 16 tanh, 400 iterations (original setting)	0.0178
PENN	2 × 16 tanh, 800 iterations	0.0178
PENN	2 × 32 tanh, 200 iterations	0.0173
PENN	2 × 32 tanh, 400 iterations	0.0174
PENN	2 × 32 tanh, 800 iterations	0.0174
Data-driven network	2 × 16 tanh, weight decay 10^{-5}	0.0623
Data-driven network	2 × 16 tanh, weight decay 10^{-3}	0.0351
Data-driven network	4 × 48 tanh, weight decay 10^{-5} (original setting)	0.0826
Data-driven network	4 × 48 tanh, weight decay 10^{-3}	0.0346

Appendix D. PINN diagnostics and training histories

Table D.1. PINN diagnostic runs (slab model, training mixes, seed 0): final loss terms, training RMSE, PDE residual, solver check and permeability bias relative to the classical slab estimates.

Run	Total loss	Data term	PDE term	Training RMSE	PDE residual RMS	Solver check R^2	Bias of log K_{int}
As in the comparison	0.135	0.113	0.0021	0.0034	0.048	0.864	−1.13
Parameters started at classical values	0.137	0.113	0.0024	0.0034	0.049	0.893	−1.08
PDE weight λ = 100	0.186	0.107	0.0007	0.0033	0.034	0.898	−1.03
Synthetic training data (known parameters)	0.020	0.001	0.0019	0.0003	0.046	0.828	−1.13
Parameters fixed at classical values	0.135	0.101	0.0033	0.0032	0.107	0.999	+0.00

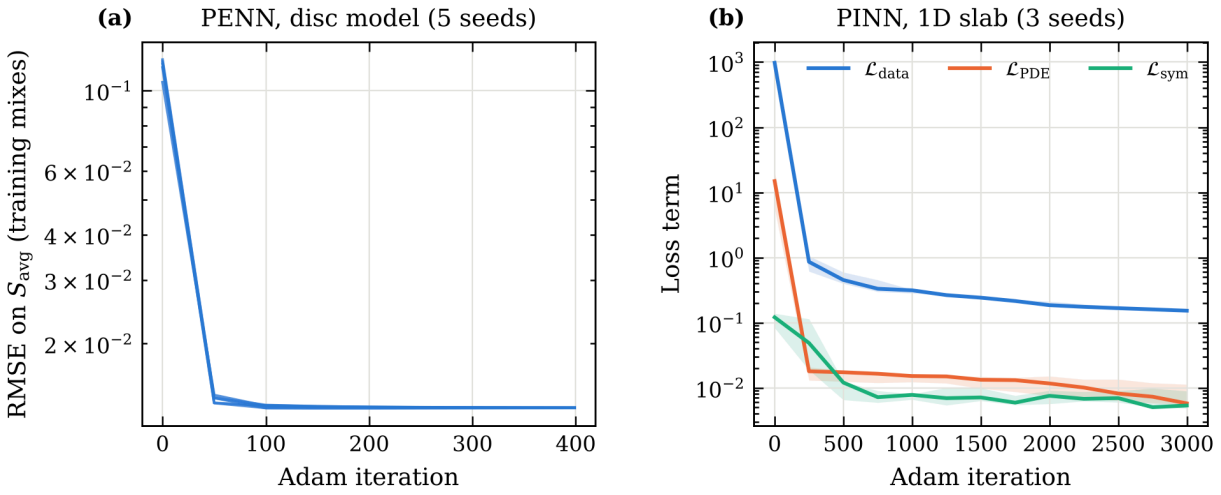


Fig. D.1. Training histories: (a) RMSE of the PENN on the training mixes (five seeds); (b) loss terms of the PINN (median and range over three seeds).

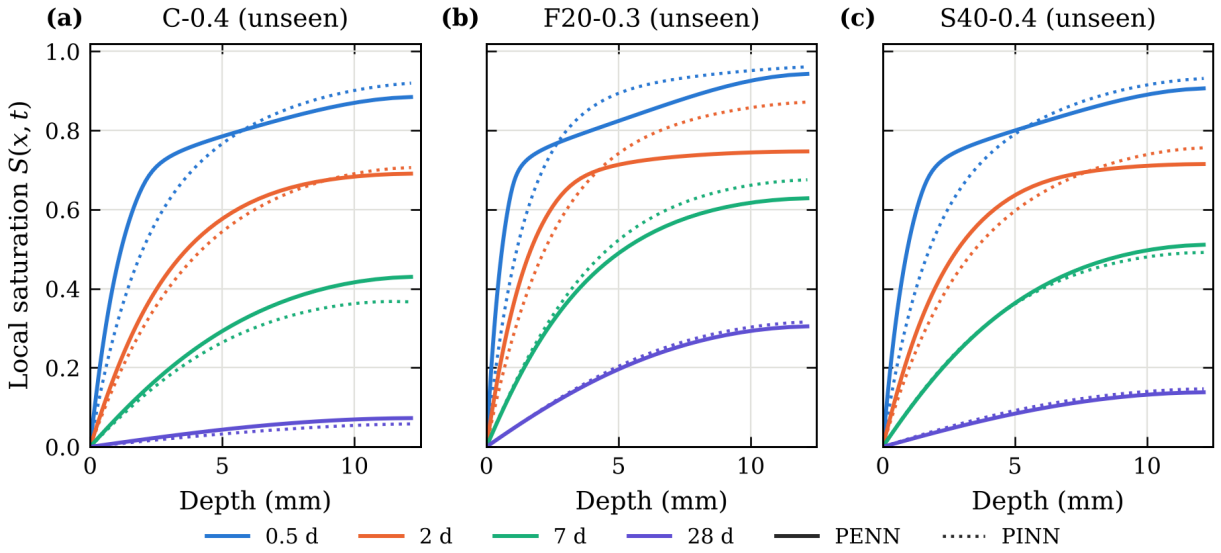


Fig. D.2. Saturation profiles of three withheld mixes on the slab model, computed by the direct solver with the PENN parameters (solid) and by the PINN (dotted). The two differ most during the first two days.

References

[1] P.K. Mehta, P.J.M. Monteiro, Concrete: Microstructure, Properties, and Materials, 4th ed., McGraw-Hill Education, New York, 2014.

[2] A.M. Neville, Properties of Concrete, 5th ed., Pearson Education, Harlow, 2011.

[3] V. Baroghel-Bouny, Water vapour sorption experiments on hardened cementitious materials. Part I: Essential tool for analysis of hygral behaviour and its relation to pore structure, Cem. Concr. Res. 37 (2007) 414–437. https://doi.org/10.1016/j.cemconres.2006.11.019

[4] V. Baroghel-Bouny, M. Mainguy, T. Lassabatère, O. Coussy, Characterization and identification of equilibrium and transfer moisture properties for ordinary and high-performance cementitious materials, Cem. Concr. Res. 29 (1999) 1225–1238.

[5] M. Mainguy, O. Coussy, V. Baroghel-Bouny, Role of air pressure in drying of weakly permeable materials, J. Eng. Mech. 127 (2001) 582–592.

[6] X. Chen, T. Sanchez, D. Conciatori, H. Chaouki, L. Sorelli, B. Selma, M. Chekired, Numerical modeling of 2D hygrothermal transport in unsaturated concrete with capillary suction, J. Build. Eng. 45 (2022) 103640. https://doi.org/10.1016/j.jobe.2021.103640

[7] C. Hall, W.D. Hoff, Water Transport in Brick, Stone and Concrete, 2nd ed., Spon Press, London, 2012.

[8] S. Zhuang, Q. Wang, M. Zhang, Water absorption behaviour of concrete: Novel experimental findings and model characterization, J. Build. Eng. 53 (2022) 104602. https://doi.org/10.1016/j.jobe.2022.104602

[9] Z.P. Bažant, L.J. Najjar, Nonlinear water diffusion in nonsaturated concrete, Matér. Constr. 5 (1972) 3–20. https://doi.org/10.1007/BF02479073

[10] H.M. Künzel, Simultaneous Heat and Moisture Transport in Building Components: One- and Two-Dimensional Calculation Using Simple Parameters, Fraunhofer IRB Verlag, Stuttgart, 1995.

[11] D. Conciatori, H. Sadouki, E. Brühwiler, Capillary suction and diffusion model for chloride ingress into concrete, Cem. Concr. Res. 38 (2008) 1401–1408.

[12] J. Kaipio, E. Somersalo, Statistical and Computational Inverse Problems, Springer, New York, 2005.

[13] A. Tarantola, Inverse Problem Theory and Methods for Model Parameter Estimation, SIAM, Philadelphia, 2005.

[14] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, J. Timmer, Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood, Bioinformatics 25 (2009) 1923–1929. https://doi.org/10.1093/bioinformatics/btp358

[15] W. Liu, G. Liu, X. Zhu, Applicability of machine learning algorithms in predicting chloride diffusion in concrete: Modeling, evaluation, and feature analysis, Case Stud. Constr. Mater. 21 (2024) e03573.

[16] M. Raissi, P. Perdikaris, G.E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, J. Comput. Phys. 378 (2019) 686–707. https://doi.org/10.1016/j.jcp.2018.10.045

[17] G.E. Karniadakis, I.G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, L. Yang, Physics-informed machine learning, Nat. Rev. Phys. 3 (2021) 422–440. https://doi.org/10.1038/s42254-021-00314-5

[18] S. Cuomo, V.S. Di Cola, F. Giampaolo, G. Rozza, M. Raissi, F. Piccialli, Scientific machine learning through physics-informed neural networks: where we are and what's next, J. Sci. Comput. 92 (2022) 88.

[19] A.G. Baydin, B.A. Pearlmutter, A.A. Radul, J.M. Siskind, Automatic differentiation in machine learning: a survey, J. Mach. Learn. Res. 18 (2018) 1–43.

[20] A.M. Tartakovsky, C.O. Marrero, P. Perdikaris, G.D. Tartakovsky, D. Barajas-Solano, Physics-informed deep neural networks for learning parameters and constitutive relationships in subsurface flow problems, Water Resour. Res. 56 (2020) e2019WR026731.

[21] Y. Wan, W. Zheng, Y. Wang, Identification of chloride diffusion coefficient in concrete using physics-informed neural networks, Constr. Build. Mater. 393 (2023) 132049. https://doi.org/10.1016/j.conbuildmat.2023.132049

[22] Y. Wan, W. Zheng, Physics-informed neural network approach for identifying free chloride diffusion coefficient in alkali-activated mortars, J. Build. Eng. 112 (2025) 113820. https://doi.org/10.1016/j.jobe.2025.113820

[23] J. Huang, J. Hu, Z. Li, Y. Zhang, Y. Cheng, Physics-informed neural network for predicting the moisture diffusion and parameter inversion in stone heritage, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci. XLVIII-M-9-2025 (2025) 595–600. https://doi.org/10.5194/isprs-archives-XLVIII-M-9-2025-595-2025

[24] J.M. Lee, C.J. Lee, Inverse estimation of moisture diffusion model for concrete using artificial neural network, Materials 15 (2022) 5945. https://doi.org/10.3390/ma15175945

[25] A.S. Krishnapriyan, A. Gholami, S. Zhe, R.M. Kirby, M.W. Mahoney, Characterizing possible failure modes in physics-informed neural networks, Adv. Neural Inf. Process. Syst. 34 (2021) 26548–26560.

[26] S. Wang, Y. Teng, P. Perdikaris, Understanding and mitigating gradient flow pathologies in physics-informed neural networks, SIAM J. Sci. Comput. 43 (2021) A3055–A3081.

[27] S. Wang, S. Sankaran, P. Perdikaris, Respecting causality for training physics-informed neural networks, Comput. Methods Appl. Mech. Eng. 421 (2024) 116813.

[28] P. Rathore, W. Lei, Z. Frangella, L. Lu, M. Udell, Challenges in training PINNs: A loss landscape perspective, in: Proc. 41st Int. Conf. Mach. Learn. (ICML), Proc. Mach. Learn. Res. 235 (2024) 42159–42191.

[29] C. Rackauckas, Y. Ma, J. Martensen, C. Warner, K. Zubov, R. Supekar, D. Skinner, A. Ramadhan, A. Edelman, Universal differential equations for scientific machine learning, arXiv:2001.04385 (2020).

[30] N. Thuerey, P. Holl, M. Mueller, P. Schnell, F. Trost, K. Um, Physics-based Deep Learning, arXiv:2109.05237 (2021).

[31] T. Bandai, T.A. Ghezzehei, P. Jiang, P. Kidger, X. Chen, C.I. Steefel, Learning constitutive relations from soil moisture data via physically constrained neural networks, Water Resour. Res. 60 (2024) e2024WR037318. https://doi.org/10.1029/2024WR037318

[32] ASTM C1585, Standard Test Method for Measurement of Rate of Absorption of Water by Hydraulic-Cement Concretes, ASTM International, West Conshohocken, PA.

[33] E. Hairer, G. Wanner, Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems, 2nd ed., Springer, Berlin, 1996.

[34] R.G. Keys, Cubic convolution interpolation for digital image processing, IEEE Trans. Acoust. Speech Signal Process. 29 (1981) 1153–1160. https://doi.org/10.1109/TASSP.1981.1163711

[35] C.E. Rasmussen, C.K.I. Williams, Gaussian Processes for Machine Learning, MIT Press, Cambridge, MA, 2006.

[36] D.P. Kingma, J. Ba, Adam: A method for stochastic optimization, in: Proc. 3rd Int. Conf. Learn. Represent. (ICLR), San Diego, 2015. arXiv:1412.6980.

[37] B. Efron, R.J. Tibshirani, An Introduction to the Bootstrap, Chapman & Hall, New York, 1993.

[38] L. Yang, X. Meng, G.E. Karniadakis, B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data, J. Comput. Phys. 425 (2021) 109913.